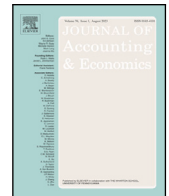




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ABSTRACT

Using novel survey data from 1426 firms in Guatemala, Honduras, and El Salvador, we examine the voluntary adoption of accounting systems by microenterprises operating in the informal economy and its association with access to credit and business growth. Despite the absence of regulatory mandates, 44 percent of informal microbusinesses maintain systematic financial records, with roughly three-quarters of those using notebooks and one-quarter using digital records. Accounting quality is closely linked to the owner's startup motivation, financial stake, and management capabilities. Informal firms that rely on larger suppliers, extend customer installment credit, engage in marketing promotions, or employ more formal organizational structures are also more likely to adopt structured recordkeeping systems, highlighting accounting's role as a managerial tool rather than as a compliance function or as a precursor to entering the formal economy. Informal firms that maintain notebooks separating business and personal accounts are significantly more likely to apply for and obtain bank loans. Our inferences for access to credit also extend to microenterprises in the Dominican Republic. In addition, accounting quality appears to support lender due diligence even among informal firms, particularly when lenders are more sophisticated. Using a broader measure of credit market status, we find that accounting quality rises monotonically across credit tiers. Finally, we find that entrepreneurs with higher-quality accounting systems are more likely to plan for business expansion. By illuminating the motivations behind accounting choices in low-enforcement settings, this study deepens our understanding of the foundational role accounting plays in early-stage business development in emerging markets.

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1. Introduction

This study examines the relationship between voluntary accounting practices and the informal economy in developing countries, where firms largely operate outside tax and business registrations, and beyond government enforcement (Ulyssea, 2020). We focus on informal microenterprises, defined as those that are not registered with the national tax authority, that are not in compliance with business registration requirements (ILO, 2020), and that employ fewer than ten persons (see European Union, ILO, IDB). Using cross-sectional data from a 2019 survey by the International Labour Organization (ILO) on 1426 microenterprises in Guatemala, Honduras, and El Salvador, we document the economic trade-offs associated with recordkeeping practices in informal firms and how these practices relate to access to credit and business growth.¹

Despite accounting for approximately 30 percent of global GDP and 50 percent of employment, the informal economy, where micro and small enterprises are dominant, remains underexplored in the accounting literature (ILO, 2020; Tomy and Wittenberg-Moerman, 2024, 2025). Their business practices are typically rudimentary (Buehn and Schneider, 2012; Halvorson-Quevedo, 1991; IFC, 2010; Ulyssea, 2020) and their financial transactions are often governed by social capital and trust-based relationships (Tomy and Wittenberg-Moerman, 2024, 2025).

Prior research on private firms shows that even absent public disclosure mandates, their accounting choices are typically shaped by tax rules, loan covenants, and statutory reporting thresholds (e.g., Bernard and Sutherland, 2025; Allee and Yohn, 2009; Ball, 2024; Beuselinck et al., 2023; Lisowsky and Minnis, 2020; Minnis, 2011). In contrast, entrepreneurs in the informal economy face no such mandates, so even maintaining rudimentary records reflects a direct cost-benefit trade-off. The contrast between formal and informal firms in our data underscores this point: nearly all formal microenterprises maintain systematic, often digitized records, whereas informal firms display wide heterogeneity.² The informal economy thus provides a distinctive lens through which to observe the foundational, emergent role of accounting, which is not readily visible in more developed and institutionally mature markets (Ball, 2024, 2025b).³

Our survey-based measures capture both whether systematic records are kept and gradations in their level of sophistication, providing direct descriptive evidence on accounting adoption in the informal sector absent regulatory mandates. We first focus on a binary outcome, the voluntary decision of informal microbusinesses to adopt a systematic recordkeeping system (*Systematic Recordkeeping*). We later exploit the survey's granularity to develop a four-level ordinal measure of *Accounting Quality*: (1) No systematic recordkeeping; (2) *Commingle NB*, where business and personal financial records are combined in a single physical, paper-based notebook; (3) *Separate NB*, where business records are maintained separately in a dedicated, paper-based notebook; and (4) *Digitized*, representing a more sophisticated accounting system (see Fig. 1). A distinctive feature of our setting is the commingled category (Jayachandran, 2020) because most microenterprise training programs emphasize separating business and personal finances (Dyer et al., 2016; McKenzie and Woodruff, 2014). While such separation may facilitate business sustainability and serve as a behavioral commitment device (Drexler et al., 2014), whether it has real effects is an open question.

In our sample of informal firms, 44 percent maintain systematic records, with nearly three-quarters relying on a paper-based notebook. Notably, more than one-quarter of these notebook users do not separate business and personal finances. In our setting, there is no separation of ownership and control, which implies that accounting system choices are less likely to reflect classic agency conflicts. Instead, these choices primarily reflect entrepreneurs' trade-offs between the costs and benefits of systematic recordkeeping. Using a reduced-form probit (ordered probit) model, we examine the association between *Systematic Recordkeeping* (*Accounting Quality*) and the determinants that shape these trade-offs.

We find that entrepreneurs with stronger business orientation (Calderon et al., 2017), but not those motivated by family considerations, and those with greater managerial human capital (Bloom et al., 2010, 2013; Bloom and Van Reenen, 2007; Karlan and Valdivia, 2011) are more likely to adopt systematic recordkeeping. Entrepreneurs who view their own equity capital or banks as desirable funding sources or who are saving toward a future capital infusion are more likely to have systematic financial records (Cassar, 2009; Dupas and Robinson, 2013). The observed association between accounting choices and owner characteristics provides novel descriptive evidence that broadens our understanding of the role of accounting in evolving business organizations (Sunder, 1997).

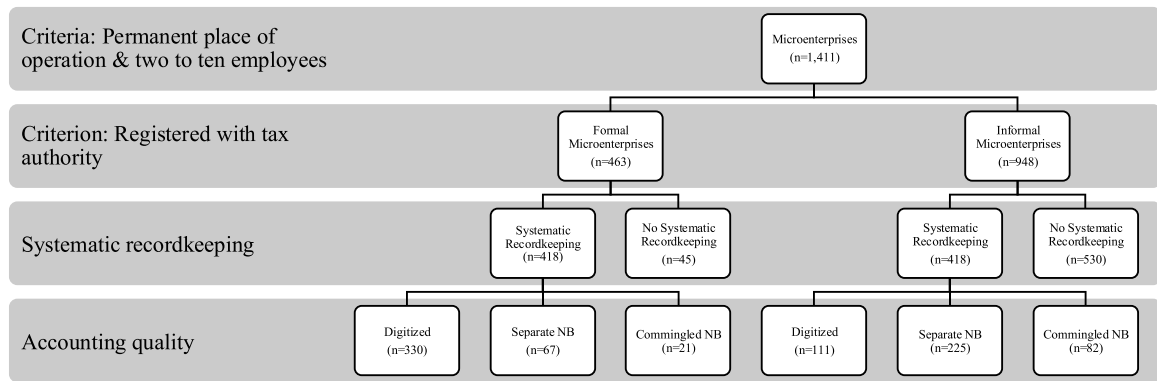
Among operational factors, firms that extend installment credit or engage in marketing promotion activities are more likely to rely on systematic financial records, with accounting plausibly serving as a managerial tool for customer acquisition and retention. Consistent with the informational demands of supply-chain relationships, informal firms that rely on large suppliers are more likely to have systematic records. Reflecting the level of informality in organizational design (Brickley et al., 2003), businesses that rely more heavily on family members or temporary workers tend to use more rudimentary records, whereas those with more full-time employees opt for higher-quality accounting.

Turning to *Accounting Quality*, we report the marginal effects from the ordered probit model for each outcome category. We find that these marginal effects indicate a shift in probability away from no systematic financial records toward higher-quality systems, with commingled notebooks playing only a limited role. For example, at the margin, firms that extend installment credit are 33

¹ We use *recordkeeping* and *accounting* interchangeably, with a slight preference for the former to avoid confusion with the more formal reporting structures associated with the latter.

² The formal firms in our sample are distinct from private firms typically studied in extant research, which face reporting and verification requirements well beyond those relevant for microenterprises in developing economies (Bernard and Sutherland, 2025).

³ For the historical evolution of recordkeeping, see Ball (2025a); Basu et al. (2009); Basu and Waymire (2006); Waymire and Basu (2008); Dickhaut et al. (2010); King and Levine (1993); and Schmandt-Besserat (1992).



The full survey sample includes 1426 firms. The figure reports the 1411 firms for which the information required to construct the accounting classifications is available.

Fig. 1. Sample description.

percentage points less likely to have no systematic records, and about 16 percentage points more likely to adopt either separate notebooks or digitized systems, relative to firms that do not extend installment credit. Robustness analyses show that accounting choices are not associated with anticipated formalization and are not merely proxies for general business sophistication. Taken together, our results align with historical evidence that merchants and money changers created accounting systems to support the efficient management of their businesses rather than as a compliance mechanism (Ball, 2025b; Basu et al., 2009).

We next examine the role of accounting in microenterprises' access to credit, a key indicator of financial development (Guiso et al., 2004). Prior research highlights the importance of social capital and trust in facilitating *supplier* credit in informal economies (Tomy and Wittenberg-Moerman, 2024). By contrast, we find that firms that prefer *bank financing* are more likely to maintain systematic financial records, raising the empirical question of whether accounting quality is associated with access to bank credit in the informal economy. For this analysis, we use the microenterprises in the formal economy as a comparison group (footnote 2).

Roughly 40 percent of the microenterprises in our sample have ever applied for a bank loan, with 34 percent of informal firms and 52 percent of formal firms doing so. Among applicants, 77 percent of informal firms and 93 percent of formal firms reported receiving the loan. To model both the decision to apply and the likelihood of approval, we estimate separate Heckman selection models for informal firms and for the benchmark sample of formal firms. Our specifications include controls for collateral, contracting reputation, customer and supplier credits, and business age (Berger and Udell, 1995; Petersen and Rajan, 1994; Banerjee et al., 2017).

For informal firms, only the coefficient on notebooks with separate accounts (*Separate NB*) is statistically significant for both loan applications and approvals, suggesting that separating business and personal financial records may be particularly relevant for credit access. For formal firms, digitized records are associated with a higher likelihood of loan applications and approvals, consistent with lenders placing greater weight on more sophisticated accounting systems. We draw similar inferences when using a broader measure of credit market status beyond mere access to bank loans, and these patterns remain robust to copula-based selection models that relax the joint normality assumption. Taken together, the results point to a positive relationship between accounting quality and credit access among microenterprises.⁴ However, we acknowledge that our cross-sectional data limit our ability to make causal inferences about the relationships we observe.

To mitigate concerns about generalizability, we replicate our findings from the credit access analysis using data from a similar ILO survey in the Dominican Republic (DR) conducted in 2023. The results also highlight the role of social capital (Tomy and Wittenberg-Moerman, 2024): small banks appear less likely to demand systematic financial records when providing small loans to informal firms, consistent with trust-based lending. For large loan requests, systematic recordkeeping is common regardless of bank type, suggesting that traditional financial due diligence becomes more important when financial stakes are larger.

Lastly, using the CA sample and controlling for the entrepreneur's human capital, firm size, and country/sector fixed effects, we find that firms with higher-quality accounting systems are more likely to report business expansion plans, particularly along the extensive margin. We also find that higher-quality accounting is associated with greater productivity (monthly sales per employee) among informal firms. Taken together, these findings shed light on the managerial role of accounting in informal firms (Datar et al., 2009).

⁴ Prior research on the role of accounting for access to credit in microenterprises is limited. For example, Najera Ruiz and Collazzo (2021) find that the use of any form of business recordkeeping by Chilean microenterprises is positively associated with prior access to credit from public or private institutions, including family and friends, and with the scale of the business, owners' age, education, and business training. Kahsay and Zeleke's (2019) study is not limited to microenterprises and it does not find a significant association between access to credit and the quality of accounting records. Apart from being peripherally related to our work, we are not aware of studies that systematically examine the antecedents and consequences of recordkeeping systems of microenterprises and their quality across the levels of formalization.

Table 1
Accounting practices definition.

Panel A: Cross tabulation of accounting categories by formalization					
Formalization	Systematic recordkeeping				Total
	No	Yes			
		Commingled NB	Separate NB	Digitized	
Informal	530 (55.9)	82 (8.6)	225 (23.7)	111 (11.7)	948 (100.0)
Formal	45 (9.7)	21 (4.5)	67 (14.5)	330 (71.3)	463 (100.0)
Total	575 (40.8)	103 (7.3)	292 (20.7)	441 (31.3)	1411 (100.0)

Panel B: Test of differences in accounting quality			
	Informal	Formal	Difference
Accounting			
Systematic Recordkeeping	0.444	0.904	−0.459***
Commingled NB	0.086	0.045	0.041***
Separate NB	0.237	0.145	0.093***
Digitized	0.117	0.713	−0.596***

The final column in Panel B reports differences in means and the results of two-sample mean t-tests, with standard errors adjusted for unequal variance between the two groups. Inferences are identical using a chi-squared test.

Because accounting evolves in parallel with broader institutional transformations, establishing causation in this setting is inherently difficult and may be unproductive in assessing its social value (Ball, 2024, 2025a,b). Using rich survey data from Central America, we provide descriptive evidence on the role of accounting outside the shadow of regulatory mandates, where adoption reflects a direct cost-benefit trade-off. This unique setting allows us to observe accounting in its foundational role as a managerial tool rather than as a compliance mechanism.⁵

The remainder of the paper is organized as follows. Section 2 examines when and why informal firms voluntarily keep systematic accounting records. Next, Section 3 explores how accounting quality relates to access to credit. Finally, Section 4 examines how accounting quality is associated with managerial decisions regarding microenterprise growth plans and productivity. Section 5 concludes.

2. When do firms in the informal economy voluntarily keep systematic accounting records?

2.1. Motivation and sample description

Public and private businesses in modern economies adopt sophisticated recordkeeping and reporting systems to participate in capital markets and to engage in explicit and implicit contracts with various stakeholders for value maximization (Armstrong et al., 2010; Beaver, 1968; Bushman and Smith, 2001; Dechow et al., 2010; Healy and Palepu, 2001; Watts and Zimmerman, 1986). If reliable records facilitate efficient exchange transactions (Ball, 2025b), how do microbusinesses operating in the informal economy adapt their accounting practices to participate effectively in the market economy? Similar to private firms in developed markets that weigh the costs and benefits of financial disclosure (Gassen and Muhn, 2024), conventional wisdom suggests that microentrepreneurs who operate outside the regulated economy are likely to adopt efficient recordkeeping methods that best meet their business needs (Allee and Yohn, 2009; Basu, 2015; Soto, 2000). In contrast to the prior research on private firms, we focus on the informal economy in developing nations to shed light on the innate demand for recordkeeping practices in businesses not driven by regulatory compulsion (Ball, 2024).

For most analyses, we use data from a single-shot survey conducted by the ILO in Central America (hereafter “CA”) in 2019. The survey was designed to ensure roughly equal representation across formalization categories (see Appendix A), countries (Guatemala, Honduras, and El Salvador), and economic sectors (retail, services, and micro-manufacturing). Data were collected through face-to-face interviews with owners of microbusinesses (defined as firms with 2 to 10 workers, including the owner), resulting in a sample of 1426 firms. Interviewers used electronic tablets to record responses in real time to minimize data-entry errors. To mitigate response bias, questions regarding formalization were asked after participants had answered all other questions, including those related to accounting records. Further details on the survey are presented in Appendix A. The median microenterprise in our sample reports monthly sales of \$780, with firms in the informal (formal) sector reporting \$640 (\$1100) (untabulated).

⁵ Given that the survey relies on a stratified, non-random sample, we re-estimate our main models using sampling weights to account for the survey design and our findings remain robust. See Internet Appendix for details.

Table 1 presents the distribution of *Systematic Recordkeeping* (Panel A) and *Accounting Quality* (Panel B) by the level of formalization among microenterprises (see Fig. 1).⁶ As expected, formal microenterprises almost universally maintain systematic records (90 percent), with most using digitized systems (71 percent) rather than notebooks (19 percent).⁷ Although informal microenterprises attempt to stay outside the legal landscape, remarkably, 44 percent have voluntarily chosen to adopt systematic recordkeeping, with roughly a quarter of these choosing to have digitized records. Among the informal firms using a notebook, more than 70 percent separate business from personal financials, indicating an innate demand for tracking business performance for sustainability or as a commitment device (Drexler et al., 2014).

2.2. Research design and model specification

Consistent with the descriptive nature of our study, we employ reduced-form models and view our evidence as purely associative rather than causal. To examine the determinants of systematic recordkeeping, we estimate the following probit model:

$$\begin{aligned} \text{Systematic Records}_i^* &= X'_{bus}\beta + X'_{owner}\gamma + X'_{fe}\delta + \epsilon_i, \text{ for } i \text{ with } \text{Formalization}_i = 0 \\ \text{Systematic Records}_i &= 1 \text{ if } \text{Systematic Records}_i^* \geq 0, \\ \text{Systematic Records}_i &= 0 \text{ if } \text{Systematic Records}_i^* < 0, \\ P(\text{Systematic Records}_i = 1 | \text{Formalization}_i = 0, X_{bus}, X_{owner}, X_{fe}) &= \Phi(X'_{bus}\beta + X'_{owner}\gamma + X'_{fe}\delta), \end{aligned} \quad (1)$$

where ϵ_i captures unobserved determinants of the latent propensity to maintain systematic records; $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution; X_{bus} , X_{owner} , and X_{fe} are vectors of observed variables proxying for cost-benefit trade-offs relating to business characteristics (*bus*), the owner-level characteristics (*owner*), and fixed effects for country and economic sectors (*fe*); and β , γ , and δ are vectors of parameters to be estimated (see Appendix B for variable definitions). Unlike research on public companies or private companies in largely developed economies, the factors we analyze are unaffected by mandated disclosure requirements, allowing us to observe recordkeeping choices that arise purely from private cost-benefit considerations.

The founder's startup motivation may influence business practices, given the significant overlap between manager and owner roles in microenterprises. We consider three motivational dimensions plausibly associated with the voluntary adoption of accounting practices (Calderon et al., 2017; Mel et al., 2010) (see Appendix B). Our first measure captures the entrepreneur's business-oriented nature or vocational interest in starting the venture (*Business Orientation*), similar in spirit to the opportunity entrepreneur measure in Calderon et al. (2017). Unlike those driven by a business motivation, accounting quality may be less important to owners who enter entrepreneurship to prioritize family or personal time (*Family Time*). We do not have a directional expectation for owners facing greater opportunity barriers (*Barriers*), given potentially offsetting effects.

In addition to motivation, we consider owners' socioeconomic characteristics: education (*Secondary School* and *College*) (Allee and Yohn, 2009), business training (*Management Training*), gender (*Female*), and age (*Age*). Extant research finds that management quality is closely linked to both business performance and the adoption of more sophisticated business practices (Bloom et al., 2010, 2013; Bloom and Van Reenen, 2007; Karlan and Valdivia, 2011; Weber and McKenzie, 2009). We use education and management training as proxies for managerial quality or human capital and predict that higher-quality managers are more likely to adopt advanced accounting systems. Controlling for managerial quality helps separate managerial attributes from other observable factors associated with accounting adoption, such as financing needs and operational complexity.⁸

Financing considerations may also be associated with greater demand for accounting information. Internal and external financing sources may impose varying requirements for accountability and perceptions of risk (Bruhn et al., 2010; Christensen and Nikolaev, 2012; Diamond, 1984; Diamond and Verrecchia, 1991; Myers and Majluf, 1984). Regarding internal financing, previous research shows that greater personal savings are associated with higher business investment and growth among microentrepreneurs (Dupas and Robinson, 2013). Even in the absence of owner-manager separation, owners may rely on accounting information as a planning and feedback tool to protect and grow their capital. We include an indicator for whether the entrepreneur views her own capital as a desirable source for future expansion (*Own Capital*) and a measure of the entrepreneur's desire to save in order to build capital for growth (*Saving for Investment*), predicting a positive marginal effect for both.

To explore the effects of external financing, we include indicators representing the desirability of three external funding sources for future business expansion identified by the entrepreneurs: informal lenders (*Informal Lenders*), microfinance (*Microfinance*), and banks (*Bank*). In the informal economy, credit extension is likely influenced by mutual trust (Tomy and Wittenberg-Moerman, 2025, 2024). As a result, we expect accounting quality to be less valuable when borrowing from informal lenders.

⁶ The distribution of microenterprises without a systematic recordkeeping system in our setting is similar in spirit to that reported in Mexico's National Survey of Microenterprises (ENAMIN), where more than half of the microenterprises state that they keep no records at all (Cotler and Woodruff, 2008).

⁷ For defining a systematic recordkeeping system, we use the following specific yes or no question in the survey: "Does this business keep formal accounting records?" The participating microentrepreneurs were not provided a specific definition of "formal accounting records". *Digitized* captures organizations using software to track financial transactions (e.g., Excel) as well as organizations that are likely to rely on digitized records through hiring an in-house accountant or engaging a professional accounting service.

⁸ In the context of public companies, extensive literature in accounting, finance, strategy, and management examines the association between CEO personal and socioeconomic characteristics (e.g., narcissism, overconfidence, hubris, optimism, gender, humility, entrepreneurial orientation, and education) and firm performance (Shen, 2021). Accounting literature also studies CEO personal attributes and financial reporting quality (Ahmed and Duellman, 2013; Schrand and Zechman, 2012).

Unlike traditional banks, microfinance lenders conduct more nuanced financial due diligence that relies on borrowers' "financial wherewithal... [and] references from customers and neighbors" (Kota, 2007). As such, whether access to microfinance depends on higher-quality accounting systems remains an empirical question. In contrast, given the extensive research on the due diligence for bank loans, we expect microenterprises that view banks as a desirable funding source to have higher accounting quality (e.g., Cassar, 2009).

Regarding operational demands, we include an indicator for whether the firm relies on large suppliers (*Large Supplier*), as a large supplier's willingness to contract is likely to depend on the quality of the customer's financial records (Naidu and Ranjeeni, 2024). Similarly, accounting quality may enable effective management of customer relationships, proxied by an indicator for firms that provide installment financing to customers (*Customer Credit*). Furthermore, extant research suggests that microenterprises benefit from an effective marketing strategy, although they learn by trial and error (Morgan, 2019). Anecdotal evidence is consistent with these firms tracking marketing activities, potentially to assess their effectiveness (Porto et al., 2017). Our empirical proxy, *Marketing Promotion*, is the sum of six indicators representing product or service promotions through local newspapers, radio, murals, flyers, word of mouth, and social media. We posit that recordkeeping could help the microentrepreneur better assess channel effectiveness and manage the marketing budget efficiently. Therefore, we predict that the likelihood of entrepreneurs adopting a systematic recordkeeping system increases in the number of promotional channels deployed. We also control for the presence of additional business establishments (*Business Complexity*) (Allee and Yohn, 2009), years of business operation (*Firm Age*), numbers of full-time (*Number of Workers*) and temporary (*Temporary Workers*) employees, and the number of family members working in the business (*Number of Family Workers*).

2.3. Main results

We begin by providing descriptive statistics on the survey data. While our primary analyses center on informal firms, Table A.1 in Appendix A compares the means of the determinants of systematic recordkeeping (Eq. (1)) between the informal and formal microenterprises. Several empirical regularities emerge. Regarding owners' motivations and characteristics, informal entrepreneurs are more likely to indicate a lack of employment opportunities as a reason for starting their business. They also tend to be female, younger, less educated, and less likely to have received business training. In terms of financing, informal firms express a stronger interest in funding from informal lenders, whereas formal firms show a greater preference for bank financing. From an operational standpoint, formal firms are roughly twice as likely to have a large supplier and more than four times as likely to extend installment credit to customers. Informal microbusinesses are less likely to engage in marketing promotion activities, have more than one business establishment, or employ more full-time and temporary workers. Overall, the comparative descriptive patterns align with economic intuition and lend credibility to the reliability and representativeness of the survey data.⁹

Table A.2 in Appendix A reports descriptive statistics for the main variables included in the accounting model. The marginal effects from estimating the accounting adoption model for informal firms using a probit specification are presented in Table 2. All major categories of determinants (owner motivations, owner characteristics, financing, and operational factors) are meaningfully associated with the voluntary choice of accounting systems. Among owner motivations, business-oriented entrepreneurs have a 0.58 higher probability of maintaining systematic recordkeeping, with no significant results for those who experience opportunity barriers or prioritize family time.¹⁰ Turning to owner characteristics, management quality plays a key role: entrepreneurs who participated in business training programs have a 16-percentage-point-higher propensity for systematic recordkeeping.

More importantly, microentrepreneurs appear to value systematic recordkeeping when personal capital is at stake. Informal entrepreneurs who plan to save for future capital are 33 percentage points more likely to maintain systematic recordkeeping, the largest marginal effect among all financing sources, followed by an 11-percentage-point increase in likelihood when they perceive their own capital as a desirable source of financing.

Among the external sources of financing, we find that accounting quality plays a differentiated role. Informal firms do not place greater emphasis on accounting systems when they view microfinance as a desirable funding source, likely because microfinance providers demand less hard information. By contrast, firms desirous of accessing bank credit are 10 percentage points more likely to have systematic financial records. While the finding is consistent with the extensive literature on accounting quality and bank loans, the association emerges even at the level of informal microbusinesses, where firms are unlikely to have audited or reviewed financial statements, or even a complete set of accruals-based financial statements. The result highlights that, even in low-enforcement environments, accounting information can complement trust-based mechanisms that typically govern credit transactions in the informal economy (Tomy and Wittenberg-Moerman, 2024).¹¹

⁹ In an untabulated analysis, we find that informal firms perceive a greater regulatory burden and higher business costs for participating in the formal economy and are more likely to view informality as a cultural norm (Mel et al., 2013; La Porta and Shleifer, 2014; Levenson and Maloney, 1998). Williams et al. (2016) find that the likelihood of formalization by microenterprises in Pakistan increases in the entrepreneur's age, education, household income, and firm age, but decreases in perceptions of regulatory burden and corruption.

¹⁰ The marginal effect of 0.584 corresponds to the differential effect between an entrepreneur who strongly agrees with all four survey items relating to business orientation (*Business Orientation* = 1) and one who strongly disagrees with all four items (*Business Orientation* = 0). In our informal firm sample, the lowest value of *Business Orientation* is 0.1875 ($n = 4$), and the highest is 1 ($n = 17$). A one standard deviation increase in *Business Orientation* (0.132) is associated with a 0.077 increase in the probability of choosing systematic recordkeeping, which we view as a more representative measure of economic significance.

¹¹ First, we posit that accounting records play a more critical role in facilitating direct access to capital from financial intermediaries, whereas trust and social capital may be more influential in accessing credit through supply chains (e.g., from trade creditors). Second, the marginal effect of *Informal Lenders* is positive and statistically significant, while the effect for *Microfinance* is statistically insignificant. However, given the limited theoretical guidance regarding either source, future research could more directly investigate how these financing channels influence the accounting quality of microenterprises.

Table 2
Determinants of systematic recordkeeping by informal firms marginal effects.

	Base model (1)	Preference model (2)	Shapley R^2 Decomp (3)
Owner motivation			
<i>Business Orientation</i>	0.584*** (4.84)	0.578*** (4.53)	0.024
<i>Family Time</i>	0.044 (0.55)	0.052 (0.61)	
<i>Barriers</i>	0.084 (0.94)	0.084 (0.87)	
Owner characteristics			
<i>Secondary School</i>	−0.051 (−1.57)	−0.054 (−1.57)	
<i>College</i>	0.078 (1.21)	0.065 (0.96)	
<i>Management Training</i>	0.160*** (5.20)	0.163*** (4.97)	0.020
<i>Female</i>	0.005 (0.16)	0.010 (0.33)	
<i>Age</i>	−0.001 (−0.58)	−0.001 (−0.62)	
Financing			
<i>Saving for Investment</i>	0.326*** (4.61)	0.302*** (4.01)	0.023
<i>Own Capital</i>	0.110*** (3.48)	0.106*** (3.12)	0.011
<i>Informal Lenders</i>	0.067** (2.12)	0.069** (2.03)	0.003
<i>Microfinance</i>	0.014 (0.34)	0.019 (0.44)	
<i>Bank</i>	0.101*** (2.93)	0.075** (2.02)	0.013
Operations			
<i>Large Supplier</i>	0.106*** (2.84)	0.126*** (3.21)	0.010
<i>Customer Credit</i>	0.358*** (3.78)	0.398*** (3.90)	0.009
<i>Marketing Promotion</i>	0.066*** (4.75)	0.067*** (4.61)	0.024
<i>Business Complexity</i>	−0.095 (−0.95)	−0.098 (−0.89)	
<i>Firm Age</i>	−0.008*** (−3.26)	−0.008*** (−3.17)	0.008
<i>Number of Family Workers</i>	−0.039** (−2.43)	−0.043** (−2.54)	0.002
<i>Temporary Workers</i>	−0.095* (−1.93)	−0.101** (−1.97)	0.003
<i>Number of Workers</i>	0.056*** (4.33)	0.049*** (3.64)	0.024
Prefer Formalization			
		0.023 (0.72)	
Observations	916	836	916
Pseudo- R^2	0.20	0.19	0.20

Table 2 presents marginal effects of probit models for the informal firms. Column (2) extends the analysis by including stated preference for formalization as an additional explanatory variable (*Prefer Formalization*). Column (3) reports the Shapley R^2 values for the significant variables from the base model (column (1)), treating the insignificant variables and fixed effects as two additional sets of variables that have a total of 0.012 and 0.018, respectively. All regressions include country and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

On the operational side, consistent with the higher informational demands of transacting with larger suppliers, informal firms are 11 percentage points more likely to maintain systematic financial records. At the other end of the value chain, firms that extend installment credit are 36 percentage points more likely to adopt systematic recordkeeping, reflecting the role of accounting systems in monitoring customer credit and facilitating access to working capital financing. Similarly, adding one marketing promotion channel increases the likelihood of systematic recordkeeping by approximately 6.6 percentage points. Therefore, even informal microbusinesses rely on accounting as a managerial tool for customer acquisition and retention. Firms with informal organizational structures that rely on family members or part-time workers are less likely to maintain systematic records, whereas those employing

more full-time workers are more likely to do so. Overall, our analysis suggests that firms that choose informality nevertheless voluntarily maintain systematic financial records, consistent with cost-benefit trade-offs in entrepreneurial decision-making.

To assess the relative contribution of individual determinants of systematic accounting adoption, we decompose model fit by averaging each regressor's marginal contribution across all possible model subsets using the Shapley-value logic from cooperative game theory (Luchman, 2021).¹² The analysis is intended to characterize the relative importance of the determinants rather than to perform model selection. The method evaluates how much each variable improves explanatory power when added to different combinations of the other regressors and averages these incremental contributions across all possible orderings, yielding a measure of each determinant's importance while accounting for correlations among the explanatory variables.

The Shapley-value R^2 decomposition of the determinants of systematic recordkeeping is reported in column (3) of Table 2 for our base model. While the marginal effects are informative, the R^2 decomposition provides complementary insights by highlighting the relative importance of broader constructs. In particular, owner business orientation, managerial human capital (management training), personal financial stake (planned savings for investment), customer engagement (marketing promotions), and firm scale (the number of workers) each contribute roughly one-tenth of the overall explanatory power of 20 percent.

What complementary attributes characterize informal firms at the tails of the accounting adoption decision? The same five variables that account for the largest shares of explanatory power in the R^2 decomposition also characterize firms at the tails of the distribution of accounting adoption probabilities (see Internet Appendix). Among firms in the highest quartile of all five variables ($n = 58$), 78 percent maintain systematic records, compared to a sample mean of 44 percent, whereas 87 percent of firms in the lowest quartile ($n = 30$) do not, compared to a sample mean of 56 percent. Overall, this evidence is consistent with systematic financial recordkeeping being uncommon among firms lacking bundles of complementary attributes.

Because our analysis centers on informal firms, an important question is whether the role of accounting is evident after conditioning on the structural demands of formalization. Estimating Table 2 probit model for formal firms, we find that only four explanatory variables are statistically significant at conventional levels compared to 13 in the informal-firm sample (untabulated). By focusing on informal firms, we shed light on the voluntary economic forces at work, providing a gateway into the foundational role of recordkeeping.

A potential concern is that informal firms may adopt systematic financial records in anticipation of future formalization, introducing correlated omitted variable bias. To mitigate this concern, we re-estimate the probit model for the informal firms after incorporating *Stated Preference for Formalization* as an additional control and report the results in column (2) of Table 2.¹³ The coefficient on this variable is not statistically significant, and the marginal effects of the other determinants remain virtually unchanged from column (1).

2.4. Gradations of accounting quality

We estimate an ordered probit model using the ordinal variable *Accounting Quality* and report the marginal effects for informal firms in Table 3. Because the predicted probabilities across accounting categories must sum to one, the marginal effects across categories sum to zero by construction, reflecting how changes in an explanatory variable redistribute probability across the accounting categories. Because the overall inferences are consistent with those from the binary model, we limit the discussion to key differences and new insights rather than providing a detailed comparison of Tables 2 and 3.

The marginal effects in Table 3 indicate systematic shifts from non-systematic recordkeeping to either of the two higher-quality systems or vice versa. Entrepreneurs who intend to save for future investment are 14 (13) percentage points more likely to have digitized records (separate books) than those with no such plans, with the marginal effects largely reflecting a shift in probability away from non-systematic recordkeeping. A similar pattern emerges on the operational side, which holds across nearly all significant variables.¹⁴ Although we find statistical significance for *Commingle NB*, the marginal effects are an order of magnitude smaller than those of the two higher-quality accounting systems. Taken together, the results in Tables 2 and 3 provide descriptive evidence on the determinants operating along both the intensive and extensive margins of accounting system choice.¹⁵

2.5. Distinguishing recordkeeping from general engagement with business technology

To ensure that our findings are not merely capturing broader technology adoption or general business sophistication, we conduct a robustness test by regressing an indicator for entrepreneurs who use a business cell phone on the same set of determinants for systematic recordkeeping. Table 4 reports marginal effects from the joint estimation of the two probit models, cell phone use

¹² We focus on the 13 variables that are statistically significant in our base model (Table 2), treating the insignificant variables and fixed effects as two additional sets of variables, yielding 32,767 ($2^{15}-1$) possible model subsets. Our inferences are identical when we always include the insignificant variables and fixed effects as determinants and focus only on the 13 variables that are statistically significant in our base model (Table 2) for determining the model subsets (8191 or ($2^{13}-1$) model subsets).

¹³ *Stated Preference for Formalization* takes a value of one when entrepreneurs choose the option "Duly registering your business" to the question "Do you think it is preferable for a microbusiness in this country to operate as a formal business or to operate as an informal business?" and zero otherwise.

¹⁴ Similar marginal effects in columns (1) and (2) of Table 3 reflect comparable covariate shifts along the latent accounting-quality dimension, even though *Digitized* and *Separate NB* have different predicted probability distributions.

¹⁵ We examine whether home-based firms are more likely to commingle personal and business finances. Descriptively, these firms are more likely to rely on non-systematic recordkeeping rather than separate or commingled notebooks, and an indicator for home-based firms is not statistically significant in our base model (Table 2).

Table 3

Determinants of the choice of accounting quality by informal firms ordered probit marginal effects.

	Digitized (1)	Separate NB (2)	Commingled NB (3)	None (4)
Owner motivation				
<i>Business Orientation</i>	0.218*** (3.54)	0.205*** (3.47)	0.030*** (3.08)	−0.453*** (−3.57)
<i>Family Time</i>	−0.013 (−0.34)	−0.012 (−0.34)	−0.002 (−0.34)	0.027 (0.34)
<i>Barriers</i>	0.015 (0.37)	0.014 (0.37)	0.002 (0.37)	−0.032 (−0.37)
Owner characteristics				
<i>Secondary School</i>	−0.019 (−1.34)	−0.018 (−1.33)	−0.003 (−1.32)	0.039 (1.34)
<i>College</i>	0.090*** (2.90)	0.084*** (2.89)	0.012** (2.57)	−0.186*** (−2.93)
<i>Management Training</i>	0.059*** (4.04)	0.055*** (4.10)	0.008*** (3.57)	−0.123*** (−4.19)
<i>Female</i>	−0.013 (−0.94)	−0.012 (−0.95)	−0.002 (−0.95)	0.026 (0.95)
<i>Age</i>	0.000 (−0.70)	0.000 (−0.70)	0.000 (−0.70)	0.001 (0.70)
Financing				
<i>Saving for Investment</i>	0.140*** (4.16)	0.132*** (4.29)	0.019*** (3.69)	−0.291*** (−4.36)
<i>Own Capital</i>	0.040*** (2.80)	0.037*** (2.75)	0.005** (2.57)	−0.083*** (−2.81)
<i>Informal Lenders</i>	0.009 (0.68)	0.009 (0.67)	0.001 (0.67)	−0.019 (−0.68)
<i>Microfinance</i>	−0.015 (−0.82)	−0.014 (−0.83)	−0.002 (−0.82)	0.031 (0.83)
<i>Bank</i>	0.053*** (3.45)	0.050*** (3.42)	0.007*** (3.01)	−0.110*** (−3.50)
Operations				
<i>Large Supplier</i>	0.062*** (3.62)	0.058*** (3.80)	0.008*** (3.29)	−0.129*** (−3.79)
<i>Customer Credit</i>	0.160*** (3.28)	0.151*** (3.32)	0.022*** (2.98)	−0.333*** (−3.36)
<i>Marketing Promotion</i>	0.027*** (5.24)	0.026*** (5.15)	0.004*** (4.01)	−0.057*** (−5.42)
<i>Business Complexity</i>	−0.030 (−0.61)	−0.028 (−0.61)	−0.004 (−0.60)	0.062 (0.61)
<i>Firm Age</i>	−0.003*** (−3.16)	−0.003*** (−3.15)	0.000*** (−2.91)	0.007*** (3.21)
<i>Number of Family Workers</i>	−0.022*** (−2.95)	−0.020*** (−2.92)	−0.003*** (−2.74)	0.045*** (2.98)
<i>Temporary Workers</i>	−0.040* (−1.82)	−0.038* (−1.81)	−0.005* (−1.76)	0.083* (1.83)
<i>Number of Workers</i>	0.030*** (5.09)	0.029*** (5.26)	0.004*** (4.12)	−0.063*** (−5.42)

“None” refers to firms without systematic financial records. See [Appendix B](#) for accounting quality variable definitions. The number of observations in the model is 910. The model is estimated using country and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

and systematic recordkeeping, using a seemingly unrelated regression (SUR) framework. We also report the p-values from tests comparing the coefficients between the models.¹⁶

First, the explanatory power of the cell phone probit model is less than half that of the systematic recordkeeping model (0.09 vs. 0.20), suggesting that different factors are associated with business cell phone use compared with the adoption of systematic recordkeeping. Second, several predictors that are significantly associated with systematic recordkeeping, such as saving for investment, a preference for own capital or bank financing, and the degree of organizational informality, do not predict cell phone use. However, consistent with intuition, owners of firms with more sophisticated value-chain relationships are more likely to use a cell phone and maintain systematic financial records. Overall, these findings suggest that systematic recordkeeping reflects underlying economic behaviors and incentives rather than general engagement with business technology or infrastructure.

¹⁶ Because we estimate a nonlinear SUR model, the slope coefficients in the base model (column (1) of [Table 2](#)) do not exactly match those reported in column (2) of [Table 4](#).

Table 4

Regression of business cell phone use by informal firms on the determinants of systematic recordkeeping.

	Cell phone (1)	Systematic recordkeeping (2)	p-value of difference (3)
Owner motivation			
<i>Business Orientation</i>	0.310** (2.48)	0.589*** (4.88)	0.064
<i>Family Time</i>	−0.077 (−0.89)	0.044 (0.54)	0.361
<i>Barriers</i>	0.018 (0.19)	0.082 (0.93)	0.601
Owner characteristics			
<i>Secondary School</i>	0.062* (1.79)	−0.050 (−1.56)	0.026
<i>College</i>	0.132* (1.95)	0.078 (1.22)	0.658
<i>Management Training</i>	0.076** (2.26)	0.161*** (5.26)	0.045
<i>Female</i>	−0.067** (−2.09)	0.005 (0.17)	0.138
<i>Age</i>	0.004** (2.17)	−0.001 (−0.61)	0.066
Financing			
<i>Saving for Investment</i>	−0.092 (−1.25)	0.324*** (4.60)	0.000
<i>Own Capital</i>	−0.044 (−1.26)	0.110*** (3.48)	0.001
<i>Informal Lenders</i>	0.061* (1.78)	0.066** (2.11)	0.787
<i>Microfinance</i>	0.048 (1.08)	0.015 (0.36)	0.630
<i>Bank</i>	−0.073* (−1.91)	0.099*** (2.91)	0.001
Operations			
<i>Large Supplier</i>	0.103** (2.56)	0.107*** (2.89)	0.772
<i>Customer Credit</i>	0.242** (2.27)	0.358*** (3.81)	0.254
<i>Marketing Promotion</i>	0.050*** (2.80)	0.066*** (4.75)	0.331
<i>Business Complexity</i>	0.009 (0.09)	−0.092 (−0.93)	0.375
<i>Firm Age</i>	−0.003 (−1.13)	−0.007*** (−3.24)	0.140
<i>Number of Family Workers</i>	0.007 (0.39)	−0.039** (−2.45)	0.063
<i>Temporary Workers</i>	0.064 (1.18)	−0.095* (−1.92)	0.043
<i>Number of Workers</i>	−0.018 (−1.35)	0.056*** (4.30)	0.000
Observations	916	916	
Pseudo- R^2	0.09	0.20	

Table 4 reports marginal effects from a Seemingly Unrelated Regression (SUR) model, jointly estimating the probit regressions of business cell phone use and systematic recordkeeping on the determinants of systematic recordkeeping for informal firms. Column (3) reports p-values from tests comparing the coefficient estimates between the two regressions. All regressions include country and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

3. Accounting quality and access to credit

3.1. Access to credit and the informal economy

Access to credit plays a vital role in economic development, acting as a conduit for investment, employment, capital accumulation, productivity growth, innovation, sustainability, and poverty alleviation.¹⁷ Building on the preceding analysis, we next examine whether the accounting quality of informal firms is associated with access to bank credit.

Guiso et al. (2004) view access to credit as a proxy for financial development and find that greater access promotes business formation, enhances competition, and fosters firm growth. However, microbusinesses, especially those in the informal sector, face substantial barriers to obtaining loans due to the lack of required documentation and the difficulty of contract enforcement (Aga and Reilly, 2011; Banerjee and Munshi, 2004; Chein and Silva, 2014). ILO (2016) suggests that lack of access to financial services is the primary obstacle faced by firms in the informal sector (Farazi, 2014; Estevão et al., 2022). These challenges often compel informal microbusinesses to rely on informal lenders charging high interest rates (Prijadi et al., 2020) or to seek alternative methods to demonstrate creditworthiness to lenders, such as leveraging social networks and soft information (Banerjee and Munshi, 2004; Berger et al., 2005; Besley and Coate, 1995).

Extensive prior research documents the role of financial reporting quality in facilitating lenders' assessment and monitoring of the credit risk of public and private firms.¹⁸ In particular, Minnis (2011) highlights the value of financial statement verification in the lending process, especially for privately held U.S. firms operating in opaque information environments. By contrast, not only are microbusinesses in the informal economy unlikely to seek third-party verification, but they also often lack established financial histories and market reputations that larger private firms might possess to access credit.¹⁹

Tomy and Wittenberg-Moerman (2025) identify three reasons why accounting information is rarely used in informal credit markets: entrenched reliance on social ties, low financial literacy, and distrust in accounting information. They examine the credit relationship between wholesalers and retailers in an informal economy setting where trade credit is the primary source of financing and access to microfinance is sparse. In contrast, we focus on the role of recordkeeping in facilitating bank credit for microenterprises in the informal economy.²⁰ The informal economy is heterogeneous, allowing both social capital/trust and information quality to play distinct roles across different segments of economic activity.

3.2. Research design and model specification

Survey respondents were asked whether they had ever applied for a bank loan, and, if so, whether the loan was approved.²¹ These two questions form the basis of our empirical analysis of credit access. Because the data are cross-sectional, we do not interpret the estimates as causal. Instead, we document patterns of association between accounting quality and loan outcomes. We do not expect loan approval to cause a microenterprise to change its accounting quality (reverse causality), but both could be jointly influenced by unobserved factors. While we cannot fully overcome the limitations of a single-shot survey, we estimate a Heckman selection model that accounts for potential endogeneity in the decision to apply for a bank loan, which could influence the loan approval decision.

We include several determinants of access to credit: whether the entrepreneur has potential collateral (*Owns Business Premise*) (Berger and Udell, 1995) or has contracting credibility (*Lessee*); whether the business receives credit from suppliers (*Supplier Credit*) or extends installment credit to customers (*Customer Credit*); the age of the business (*Firm Age*); and controls for the entrepreneur's education, gender, age, and motivations, as well as country and sector fixed effects. Prior research finds that small businesses rely more on trade credit when bank financing is constrained, but may also extend trade credit when bank financing is accessible, sometimes pledging receivables as collateral (Berger and Udell, 1995; Petersen and Rajan, 1994). A caveat is that although the survey is quite detailed, it does not capture all potential determinants of credit access for microenterprises (e.g., close ties with a few lenders) (Petersen and Rajan, 1994).

We initially include an instrument in the first stage (loan application) for identification; however, we report below a sensitivity analysis that does not require the identification of an instrument subject to the exclusion restriction (Altonji et al., 2005). Our instrument in the selection model, *I Am Never Short of Money*, takes a value of one when the entrepreneur responds, "I am never short

¹⁷ For example, Banerjee and Duflo (2014); Beck et al. (2000); Beck et al. (2007, 2008); Brown and Earle (2017) and Ayyagari et al. (2010). See also Bruhn et al. (2017) for a literature review and a methodology for estimating the finance gap faced by micro, small, and medium businesses in emerging economies. They find 81 percent unmet financing demand for microenterprises in developing countries. See Capasso et al. (2022) for a literature review on financial development and informality. See also Augsburg et al. (2015); Bari et al. (2021); Crépon et al. (2015) and Demirci-Kunt et al. (2011).

¹⁸ See Ball et al. (2008); Carrizosa and Ryan (2017); Christensen et al. (2016); Costello and Wittenberg-Moerman (2011); Dou (2020); Dyreng et al. (2017); Hope et al. (2011); Hope and Vyas (2017); Hope et al. (2009); Minnis (2011) and Minnis and Sutherland (2017).

¹⁹ Based on survey data from the U.S. Federal Reserve on small businesses with fewer than 500 employees, Cassar et al. (2015) find that higher third-party credit scores, but not the use of accrual accounting, lower the likelihood of loan denial. Unlike our study, Cassar et al. (2015) focus on the role of accrual accounting in reducing information asymmetries between borrowers and lenders of small businesses that are part of the formal sector.

²⁰ We do not view access to credit as a panacea given that prior research is ambivalent regarding a positive marginal effect of access to credit on business expansion and performance (Angelucci et al., 2015; Banerjee et al., 2015) and also finds a selection into borrowing by entrepreneurs with high returns to investment (Beaman et al., 2023).

²¹ Our correspondence with bank officers in El Salvador suggests that even informal firms can borrow despite their legal status and are typically evaluated using alternative documentation, such as lease contracts, electricity bills, purchase invoices, and sales records.

of money for regular business operations” to the question “When you are short of money for the regular business operation to whom do you turn to borrow money?” The indicator should be significantly negatively associated with the likelihood of applying for a loan (relevance), but is unlikely to determine the bank’s loan approval (exogenous). As discussed earlier, we use the microenterprises in the formal economy as a comparison group for this analysis, focusing on digitized accounting adopted by the substantial majority of these firms.

3.3. Main results

Frequency tabulations in Panel A of Table 5 indicate that formalization does not appear to be a precondition for access to bank credit: a nontrivial percentage of informal firms applied for a loan (34 percent), and the majority received approval (77 percent). However, consistent with prior research on credit constraints (e.g., Wellalage and Locke, 2016), the likelihood of applying for a loan and of getting approval both increase with formalization. Although our application rates align with previous research demonstrating low credit take-up rates (Banerjee et al., 2015; McKenzie, 2010; Pearlman, 2014), it is important to recognize that the high level of approval rates may not generalize to all developing countries, where informal businesses could face stricter barriers and have little to no access to formal credit.

In Panel B of Table 5, we report the marginal effects of the accounting quality indicators and other determinants from the separate Heckman selection models for informal and formal firms. Given that more than 70 percent of formal firms choose digitized records (Table 1) with limited variations in the other categories, we benchmark *Digitized* against all other accounting gradations for these firms. In the informal sample, we benchmark all three accounting quality gradations against the absence of systematic recordkeeping.

The instrumental variable (*I Am Never Short of Money*) has the expected negative slope in the selection equation and appears to satisfy the relevance criterion (p -value < 0.01) in both samples. We initially focus on formal firms to set a baseline for our primary sample of interest: informal firms. Firms in the formal sector with a higher quality recordkeeping system (*Digitized*) have a 0.18 higher probability of applying for a loan (column (4) of Panel B) and a 0.19 higher probability of getting loan approval (column (5) of Panel B) than those having an inferior accounting system. Their predominant choice of a high-quality accounting system is likely made to support the credit needs of firms in the formal economy.

Turning to informal firms, *Digitized* is statistically insignificant in the selection and outcome models, consistent with its low incidence and low estimation efficiency. However, *Separate NB* is statistically and economically significant, with a 9 (14)-percentage-point-greater likelihood of requesting a loan (receiving loan approval) among informal firms. Panel A of Table 1 shows that about 44 percent of informal firms maintain systematic financial records, and more than half use a separate notebook to track business records. Despite lacking a legal mandate, these microentrepreneurs appear to recognize the value of recordkeeping and voluntarily choose an accounting quality that meets their financing needs.

3.4. Robustness checks and extensions

3.4.1. Sensitivity analyses for endogeneity

Following Altonji et al. (2005), we assume that the correlation between the unobservables in the outcome and selection equations (ρ) is not directly identified, or the degree of sample selection is unknown. To assess the sensitivity of our inferences, we vary the value of ρ between 0 and 0.9 to examine its impact on our inferences relating to accounting quality in the outcome equation (Altonji et al., 2005).²² We do not report the robustness results from the selection or first-stage equation, as minor variations in its slope estimates and p -values are due to maximum likelihood estimation and do not reflect any selection bias.

The results in Panel C of Table 5 indicate that the tenor of our findings for the relevant accounting quality indicators remains unchanged. When we set $\rho = 0$, and thus impose exogeneity, the coefficient on *Separate NB* is statistically insignificant in the informal sample, while the coefficient on *Digitized* is significant at the 10 percent level in the formal sample. However, we expect a nontrivial positive correlation between the two sets of unobservables as in Altonji et al. (2005) (see “Estimated ρ ” in Panel B). Consistent with our expectation, we find that the significance of the second-stage accounting variables increases in ρ , suggesting that selection on unobservables is important in our setting. Overall, these results provide some comfort that our inferences are unlikely to be impacted by selection bias.

3.4.2. External validity: Dominican Republic sample

To mitigate concerns of generalizability, we examine the role of accounting quality in credit access using a 2023 survey administered by the ILO in the DR. Unlike in the CA sample, nearly all DR firms that requested a loan were approved, leaving insufficient variation in the loan approval status to estimate a selection model. However, the DR data include information on the loan amount received, which we use as an alternative outcome measure.²³ Panel A of Table 6 provides descriptive statistics on loan requests and loan amounts by the level of formalization. In both subsamples, we observe a high incidence of loan amounts greater than or equal to 100,000 Dominican pesos (DOP) (\$1822), so we define a *High/Low Loan* indicator as the second outcome measure based on whether the loan amount meets or exceeds this threshold.

²² Aobdia (2019) follows Altonji et al. (2005) and varies the ρ between 0 and 0.5, and Choudhary et al. (2022) choose a ρ of 0.5, which is an indication of substantial selection bias in their context, and examine the effect of sample selection on their results.

²³ Specifically, there are seven bins of loan amount ranges. For the lowest bin, we take the midpoint of zero and the upper bound of 10,000 Dominican pesos, and for the highest bin, we take the lower bound of 100,000 Dominican pesos as the loan amount. For all other bins, we take the midpoint as the loan amount. For the 10 cases in which the loan request was rejected, we set the loan amount to zero.

Table 5

Accounting quality and access to credit.

Panel A: Loan requests and loan approvals by formalization						
	Formalization					
	Informal		Formal		Total	
	No.	%	No.	%	No.	%
<i>Loan requests</i>						
No	627	65.8	224	48.0	851	59.9
Yes	326	34.2	243	52.0	569	40.1
Total	953	100.0	467	100.0	1420	100.0
<i>Loan approvals</i>						
No	76	23.3	16	6.6	92	16.2
Yes	250	76.7	227	93.4	477	83.8
Total	326	100.0	243	100.0	569	100.0
Panel B: Marginal effects from selection models for loans						
	Informal sample			Formal sample		
	Requests (Heckman)	Approvals (Heckman)	Approvals (Copula)	Requests (Heckman)	Approvals (Heckman)	Approvals (Copula)
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Commingled NB</i>	−0.014 (−0.23)	−0.075 (−0.76)	−0.063 (−0.58)			
<i>Separate NB</i>	0.087** (2.39)	0.144** (2.20)	0.170** (2.31)			
<i>Digitized</i>	−0.003 (−0.05)	0.146 (1.47)	0.172 (1.61)	0.178*** (3.53)	0.189** (2.14)	0.172** (2.51)
<i>Owns Business Premise</i>	−0.085** (−2.18)	0.080 (1.05)	0.068 (0.72)	0.063 (1.20)	0.172*** (2.78)	0.155*** (2.88)
<i>Lessee</i>	0.051 (1.27)	0.161** (2.28)	0.171** (1.96)	0.147** (2.45)	0.254** (2.36)	0.255*** (2.92)
<i>Suppliers Credit</i>	0.009 (0.10)	−0.118 (−0.71)	−0.138 (−0.79)	0.046 (0.44)	−0.159 (−1.43)	−0.113 (−1.15)
<i>Customer Credit</i>	0.152 (1.54)	0.111 (0.71)	0.134 (0.83)	0.263*** (3.37)	0.334*** (3.08)	0.326*** (3.78)
<i>Business Orientation</i>	−0.061 (−0.49)	−0.131 (−0.61)	−0.136 (−0.53)	0.100 (0.66)	0.254 (1.49)	0.226 (1.50)
<i>Family Time</i>	0.033 (0.39)	0.299** (2.03)	0.357** (2.23)	0.070 (0.64)	0.154 (1.11)	0.137 (1.09)
<i>Barriers</i>	0.238*** (2.63)	−0.168 (−0.93)	−0.178 (−0.95)	0.265** (2.03)	−0.233 (−0.80)	−0.182 (−0.73)
<i>Secondary School</i>	0.007 (0.19)	−0.023 (−0.39)	−0.021 (−0.30)	0.030 (0.55)	0.034 (0.43)	0.036 (0.52)
<i>College</i>	0.055 (0.88)	−0.029 (−0.26)	−0.036 (−0.30)	0.150** (2.15)	0.120 (1.00)	0.116 (1.23)
<i>Management Training</i>	0.062* (1.89)	0.056 (0.93)	0.074 (1.05)	0.021 (0.41)	−0.038 (−0.54)	−0.025 (−0.38)
<i>Female</i>	−0.030 (−0.97)	0.001 (0.02)	−0.008 (−0.13)	−0.027 (−0.60)	0.074 (1.13)	0.066 (1.17)
<i>Age</i>	0.002 (0.99)	−0.001 (−0.18)	0.000 (0.04)	0.004* (1.89)	0.010*** (2.60)	0.010*** (3.40)
<i>Firm Age</i>	0.006*** (2.78)	0.006 (1.48)	0.006 (1.40)	0.003 (1.06)	−0.002 (−0.32)	−0.001 (−0.13)
<i>I Am Never Short of Money</i>	−0.219*** (−6.40)			−0.231*** (−4.84)		
Observations	914	313	313	448	234	234
Estimated ρ		0.58			0.87	

Panel C: Marginal effects of outcome equation for Heckman models by selection bias

	Panel B		Fixed ρ			
			0.0	0.3	0.6	0.9
Informal firms						
<i>Commingled NB</i>	−0.075 (−0.76)	−0.067 (−0.72)	−0.076 (−0.75)	−0.077 (−0.78)	−0.063 (−0.86)	
<i>Separate NB</i>	0.144**	0.095	0.124*	0.140**	0.125***	

(continued on next page)

Table 5 (continued).

	(2.20)	(1.57)	(1.86)	(2.19)	(2.66)
<i>Digitized</i>	0.146	0.137	0.144	0.127	0.065
	(1.47)	(1.53)	(1.46)	(1.33)	(0.96)
Formal firms					
<i>Digitized</i>	0.189**	0.065*	0.097**	0.138**	0.185***
	(2.14)	(1.66)	(1.96)	(2.34)	(3.08)

Panel A reports loan request and approval rates for informal and formal firms. Panel B presents marginal effects from the standard Heckman-probit selection model for informal (formal) firms in columns 1–2 (4–5). In Panel B, all regressions also include country and sector fixed effects, and ρ is the correlation between the errors of the selection and outcome equations. In columns (3) and (6) of Panel B, we report estimates from a copula-based binary-outcome sample-selection model that nests the Heckman model and allows flexible dependence between selection and outcome errors. Panel C reports marginal effects for the Panel B Heckman models for the loan approval (column (2) and (5)), varying the correlation parameter ρ to assess robustness across different levels of selection bias. All regressions in Panel C include controls, and country and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

Table 6

Dominican Republic: Accounting quality and access to credit.

Panel A: Loan amount by formalization				
	Informal	Formal	Total	
<i>Loan not requested</i>	436 (65.3)	170 (50.9)	606 (60.5)	
<i>Loan amount < 100,000 Dominican pesos</i>	144 (21.6)	54 (16.2)	198 (19.8)	
<i>Loan amount ≥ 100,000 Dominican pesos</i>	88 (13.2)	110 (32.9)	198 (19.8)	
Total	668 (100.0)	334 (100.0)	1002 (100.0)	
Panel B: Accounting categories by formalization				
	Informal	Formal	Total	
<i>No Systematic Recordkeeping</i>	423 (64.0)	40 (12.0)	463 (46.5)	
<i>Notebook</i>	80 (12.1)	22 (6.6)	102 (10.3)	
<i>Digitized</i>	158 (23.9)	272 (81.4)	430 (43.2)	
Total	661 (100.0)	334 (100.0)	995 (100.0)	
Panel C: Marginal effects from selection models for loans for informal firms				
	Requests (1)	High/Low (Heckman) (2)	High/Low (Copula) (3)	Amount (Heckman) (4)
<i>Notebook</i>	0.085* (1.68)	0.046 (0.48)	0.052 (0.49)	−6973.207 (−0.94)
<i>Digitized</i>	0.034 (0.81)	0.240*** (3.22)	0.269*** (3.36)	16 089.635*** (3.42)
Observations	650	224	224	224

Panel A presents the distribution of loan amounts by formalization status. For loan amounts less than 100,000 Dominican pesos, there are 7 observations in the informal and 3 in the formal subsample where loans were not approved; in those cases, the loan amount is recorded as zero. Panel B reports the distribution of accounting categories by formalization, where, due to sparsity of data, we did not split *Notebook* by commingled vs. separate. Panel C reports the marginal effects of accounting quality on both the likelihood of requesting a loan and the loan amount obtained for informal firms. In column (3) of Panel C, we report estimates from a copula-based binary-outcome (High Loan) sample-selection model that nests the Heckman-probit and allows flexible dependence between selection and outcome errors. We replicated the Heckman model for the sample of formal firms using both dependent variables (High Loan and Loan Amount). However, the model does not converge for Loan Amount, and the estimated correlation between selection and outcome equation errors reached a boundary value, suggesting numerical instability or weak identification. All regressions include controls and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

In the DR sample, data sparsity leads us to combine commingled and separate categories into a single *Notebook* category (Panel B, Table 6). The overall patterns are consistent with those observed in the CA data: about 81 percent of formal firms adopt the highest level of accounting quality, compared to only 24 percent of informal firms. The Heckman selection model results for the

Table 7

Dominican Republic: Accounting quality and bank sophistication.

$$\text{Systematic records}_i = \alpha + \beta_1 \text{Low Loan}_i \times \text{Large Bank}_i + \beta_2 \text{High Loan}_i \times \text{Small Bank}_i + \beta_3 \text{High Loan}_i \times \text{Large Bank}_i + \epsilon_i$$

	Informal sample (1)	Formal sample (2)
<i>Low Loan</i> × <i>Large Bank</i>	0.269*** (3.97)	0.116 (0.64)
<i>High Loan</i> × <i>Small Bank</i>	0.463*** (3.91)	0.195 (1.00)
<i>High Loan</i> × <i>Large Bank</i>	0.422*** (5.45)	0.195 (1.11)
<i>Intercept</i>	0.109** (2.35)	0.714*** (4.13)
Test $\beta_3 - \beta_2$	−0.041	0.000
Test $\beta_3 - \beta_1$	0.153*	0.079
Observations	231	164
R^2	0.10	0.02

Table 7 examines whether firms that received high or low loans from small or large banks differ in their likelihood of having systematic recordkeeping. Column 1 (2) reports coefficients for informal (formal) firms from OLS regressions where the dependent variable is an indicator for systematic recordkeeping. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

informal firms are reported in Panel C of [Table 6](#).²⁴ While we find no statistically significant results for the loan application, informal firms with digitized records are 24 percentage points more likely to obtain a high loan (column (2)) and receive, on average, 16,089 DOP (\$293) more than other informal firms (column (4)). Overall, the association between accounting quality and microenterprises' credit access also emerges in the DR sample and is not limited to the original sample setting.²⁵

3.4.3. The role of accounting quality in a world of social capital and trust

Using the DR sample, we next examine whether accounting quality matters in an informal economy where social capital and trust may play a central role in financial transactions ([Tomy and Wittenberg-Moerman, 2024](#)). [Table 7](#) reports separate OLS regressions for informal and formal firms, with *Systematic Recordkeeping* as the dependent variable and interactions between loan size (high versus low) and bank type (small versus large) as the regressors, using small loans from small banks as the baseline.

In the informal sample, entrepreneurs applying for a small loan from a small bank have only an 11-percent probability of maintaining systematic financial records. However, for small loans, large banks are 27 percentage points more likely than small banks to lend to informal firms that have systematic financial records, suggesting that larger banks impose a higher baseline expectation for accounting quality. We surmise that small banks' comfort with lower accounting quality may be balanced by increased reliance on social capital and trust.

For large loans, the likelihood of systematic financial records is over 50 percent (slope plus the intercept) and statistically indistinguishable between small and large banks. This pattern suggests that when the financial stakes are high, traditional due diligence considerations dominate informal, trust-based mechanisms. Note that our results are largely confined to informal firms with a high propensity for systematic financial records for all combinations of bank sophistication and loan size in the formal sample (column (2) of [Table 7](#)). Future research can examine the relative roles of social versus financial due diligence in shaping credit access in informal economies.

3.4.4. Credit market status and accounting quality

So far, we have focused on access to bank loans, but credit constraints among microentrepreneurs are often broader and more nuanced, extending beyond observable loan applications. Extant research suggests that such credit constraints may stem not only from supply-side barriers but also from demand-side considerations such as perceptions of affordability, risk aversion, behavioral biases, social stigma, and financial literacy.

²⁴ We replicated the Heckman model for the sample of formal firms using both dependent variables (*High/Low Loan* and *Loan Amount*). However, the model does not converge for *Loan Amount*, and the estimated correlation between selection and outcome equation errors reached a boundary value for *Loan High*, suggesting numerical instability or weak identification.

²⁵ Among informal firms in the DR sample, we find that regardless of the loan application path (through the business entity versus in a personal capacity), microbusinesses with digitized financial records are more likely to be approved for a larger loan.

Table 8
Accounting quality and credit market status.

Panel A: Definition of credit market status				
	Needed financing	Loan requested	Loan obtained	
<i>Obtained loan</i>	–	Yes	Yes	
<i>Financing not needed</i>	No	No	–	
<i>Loan not approved</i>	–	Yes	No	
<i>Needed financing but loan not requested</i>	Yes	No	–	
Panel B: Cross tabulation of credit status and the level of formalization				
	Informal	Formal	Total	
<i>Financing needed loan not requested</i>	398 (41.8)	111 (23.8)	509 (35.8)	
<i>Loan not approved</i>	76 (8.0)	16 (3.4)	92 (6.5)	
<i>Financing not needed</i>	229 (24.0)	113 (24.2)	342 (24.1)	
<i>Obtained loan</i>	250 (26.2)	227 (48.6)	477 (33.6)	
Total	953 (100.0)	467 (100.0)	1420 (100.0)	
Panel C: Marginal effects from ordered probit				
	Obtained Loan (1)	Financing not needed (2)	Loan not approved (3)	Financing needed loan not requested (4)
Informal firms				
<i>Commingled NB</i>	–0.017 (–0.37)	–0.004 (–0.37)	0.000 (0.37)	0.021 (0.37)
<i>Separate NB</i>	0.089*** (2.86)	0.019*** (2.76)	–0.002** (–2.01)	–0.106*** (–2.89)
<i>Digitized</i>	0.075** (2.06)	0.016* (1.96)	–0.002* (–1.71)	–0.090** (–2.06)
Formal firms				
<i>Digitized</i>	0.192*** (4.15)	–0.033*** (–3.21)	–0.011*** (–2.87)	–0.147*** (–4.09)

Table 8 is based on the CA sample. Panel A defines the classification of credit market status. Panel B provides cross-tabulations of credit market status by formalization. Panel C presents marginal effects for credit market status determinants using an ordered probit estimation, separately for informal and formal firms. All regressions in Panel C include controls, and country and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

To better characterize the financial circumstances of these business owners, we construct a four-category measure of *Credit Market Status* for the CA sample: (1) obtained loan, (2) sufficient credit, (3) loan not approved, and (4) needed financing but loan not requested (Panel A of Table 8). In addition to capturing a broader notion of credit constraints, this measure allows us to move beyond the standard loan selection framework. A common limitation of loan-based selection models is that outcomes are undefined for firms that do not apply for credit. In contrast, *Credit Market Status* spans the full sample and permits estimation of a reduced-form ordered probit model that characterizes entrepreneurs' position in the credit market without conditioning on loan application behavior.

In Panel B, we find that more than 40 percent of firms in the informal sector needed financing but did not apply for a loan, compared to about 24 percent in the formal sector. Conversely, while roughly half of formal firms obtained a loan, only about 26 percent of the informal firms did so. Thus, although financial constraints are more severe for informal firms, all microenterprises face nontrivial credit frictions.

Using an ordered probit model (Panel C), we find that accounting quality varies systematically with the level of *Credit Market Status* in both formal and informal firms, suggesting a relationship that extends beyond mere access to bank loans. Moreover, both informal and formal firms that needed financing but did not apply for a bank loan (obtained one) have the lowest (highest) accounting quality. We replicate this analysis for the Dominican Republic (Table 9) and reach similar inferences for informal firms. Overall, accounting quality is associated with a broader measure of credit market status that goes beyond merely capturing access to bank loans.

3.4.5. Relaxing the joint normality assumption

The Heckman selection model used in Tables 5 and 6 relies on the assumption that the error terms in the selection (loan request) and outcome (loan approval or loan amount) equations follow a joint normal distribution. If this assumption is violated, the model may perform poorly and yield biased inferences (Hasebe, 2022). We implement an alternative sample-selection framework that allows for more flexible dependence structures between the unobserved determinants of loan application and loan outcomes.

Table 9
Dominican Republic: Accounting quality and credit market status.

Panel A: Cross tabulation of credit market status and the level of formalization			
	Informal	Formal	Total
<i>Financing needed loan not requested</i>	109 (16.3)	40 (12.0)	149 (14.9)
<i>Financing not needed</i>	327 (49.0)	130 (38.9)	457 (45.6)
<i>Low Loan</i>	144 (21.6)	54 (16.2)	198 (19.8)
<i>High Loan</i>	88 (13.2)	110 (32.9)	198 (19.8)
Total	668 (100.0)	334 (100.0)	1002 (100.0)

Panel B: Marginal effects from ordered probit models				
	High Loan (1)	Low Loan (2)	Financing not needed (3)	Financing needed not requested (4)
Informal firms				
<i>Notebook</i>	0.053** (2.05)	0.039** (2.07)	−0.031** (−2.04)	−0.061** (−2.04)
<i>Digitized</i>	0.046** (2.09)	0.034** (2.18)	−0.027** (−2.14)	−0.053** (−2.09)
Formal firms				
<i>Notebook</i>	−0.100 (−1.24)	−0.010 (−1.20)	0.054 (1.22)	0.056 (1.26)
<i>Digitized</i>	0.013 (0.20)	0.001 (0.20)	−0.007 (−0.20)	−0.007 (−0.20)

Table 9 is based on the DR sample. Panel A provides cross-tabulations of credit market status by formalization. Unlike in Central America, in the Dominican Republic sample we do not include motivations to start the business in the ordered probit models as explanatory variables, as they have a large number of missing values. Panel B presents marginal effects for credit market status determinants using an ordered probit estimation, separately for informal and formal firms. All regressions in Panel B include controls, and country and sector fixed effects. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

Specifically, we estimate a binary-outcome sample-selection model using the copula-based maximum-likelihood approach developed by Dancer et al. (2008), as implemented by Hasebe (2022), which includes the conventional Heckman-probit model as a special case.²⁶ Following Hasebe (2022), we select the copula specification that maximizes the log-likelihood among a set of candidate copulas. We report the copula-based results for the second-stage model using the Central American (Dominican Republic) sample in Panel B of Table 5 (Panel C of Table 6). The results are robust to alternative copula specifications, indicating that our conclusions do not hinge on the joint normality assumption.

4. Does accounting quality affect the intensive and extensive margins of informal microenterprise growth plans?

A large body of extant research shows that high-quality financial reporting and robust accounting practices enhance firms' investment efficiency and innovation (e.g., Biddle and Hilary, 2006; Biddle et al., 2009; Laux and Ray, 2020; Roychowdhury et al., 2019). Shroff (2017) further shows that firms' information sets change when they adopt new GAAP requirements, with real effects on their investment decisions (see Ferracuti and Stubben, 2019).

Whether these findings generalize to the rudimentary recordkeeping systems of microenterprises in the informal economy is an open question. To shed light on this issue, we test whether entrepreneurs' plans for business expansion are systematically related to the quality of their recordkeeping system, i.e., whether higher-quality information systems are associated with greater confidence in business prospects.

Although our data come from a single-shot survey, the survey elicits the entrepreneur's anticipated business development trajectories over the next five years at the time of the survey. Using the CA sample, we construct an intensive-margin measure as the sum of four indicator variables related to the expansion plans for the current business (see Appendix B), and an extensive-margin indicator based on entrepreneurs' plans to open a new commercial establishment in a different line of business. We sum the intensive- and extensive-margin measures to capture the full set of planned business expansion activities.²⁷

²⁶ In this framework, a copula combines marginal distributions of individual variables into a joint distribution by modeling their dependence structure. Selecting a copula amounts to choosing a functional form for this dependence, conditional on the specified marginals.

²⁷ We did not replicate this analysis for the DR sample, as the survey questions regarding business expansion plans have a different structure and do not include a clear distinction between the intensive and extensive margins.

Table 10
Business expansion plans by informal firms.

	Productivity (1)	Business expansion (2)	Intensive margin (3)	Extensive margin (4)
<i>Commingle</i> NB	25.000 (1.22)	0.168 (0.91)	0.086 (0.57)	0.082 (1.43)
<i>Separate</i> NB	80.417*** (3.52)	0.019 (0.14)	−0.077 (−0.68)	0.096** (2.48)
<i>Digitized</i>	100.000*** (2.60)	0.357** (2.12)	0.222 (1.62)	0.135** (2.54)
<i>Secondary School</i>	−16.250 (−1.10)	0.454*** (3.90)	0.390*** (4.03)	0.063* (1.95)
<i>College</i>	−16.250 (−0.54)	1.023*** (4.86)	0.800*** (4.76)	0.223*** (3.22)
<i>Management Training</i>	−16.667 (−1.13)	0.199* (1.68)	0.199** (2.03)	0.000 (0.00)
<i>Log of Sales</i>		−0.009 (−0.16)	0.010 (0.21)	−0.020 (−1.06)
Observations	940	940	940	940
R ²	0.03	0.08	0.08	0.05
Country and sector fixed effects	Yes	Yes	Yes	Yes

Column (1) reports quantile regression results with sales per employee as the dependent variable. Column (3) captures the intensive margin, where the dependent variable is the total number of expansion plans within the existing business, defined as the sum of indicators for sales growth, opening an additional location, hiring additional workers, and introducing new products, all in the same line of business. Column (4) captures the extensive margin, with the dependent variable equal to one if the entrepreneur plans to start a new business distinct from the current one. Column (2) uses the sum of the intensive- and extensive-margin measures from columns (3) and (4). Regressions in columns (2)–(4) are estimated using OLS. This analysis is not replicated for the Dominican Republic because the corresponding survey questions differ in structure and do not allow a clear distinction between intensive and extensive margins. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

Following the emphasis in the development economics literature on firm productivity (Bloom and Van Reenen, 2007; Drexler et al., 2014; McKenzie and Woodruff, 2017), we first present descriptive evidence on the association between firm productivity (monthly sales per employee) and accounting quality. As reported in column (1) of Table 10, informal firms that maintain separate books (digitized records) have approximately \$80 (\$100) higher productivity after controlling for the entrepreneur's human capital (*Education and Management Training*) and sector and country fixed effects, broadly consistent with the extant literature.

In columns (2)–(4) of Table 10, we report the results of regressing the business expansion plan variables on the accounting quality indicators, including the same set of controls and current business performance (*Log of Sales*). We find that informal firms with higher-quality accounting systems are more likely to have business expansion plans, with the results largely driven by the extensive margin.²⁸ Informal firms with digitized records (separate notebooks) are 42 (30) percent more likely than the sample mean to report plans to start a new business.²⁹ Because these outcomes reflect forward-looking expansion plans, the analysis is less susceptible to contemporaneous measurement effects noted in prior work (Drexler et al., 2014), although it does not speak to realized performance captured by current productivity. While prior research in the development economics literature documents associations between recordkeeping practices and sales growth in small firms, it has not examined the role of granularity and differential levels of accounting quality (McKenzie and Woodruff, 2017). Taken together, the productivity and expansion analyses provide complementary evidence on the potential role of accounting practices in supporting business sustainability.

We further explore heterogeneity in business expansion plans by entrepreneurs' business orientation, focusing on family business tradition. Family tradition is commonly viewed as a channel for intergenerational transmission of managerial human capital, but it limits reliance on formal management practices (Bloom and Van Reenen, 2007; McKenzie and Woodruff, 2017). An open question, therefore, is whether entrepreneurs with a family business background rely differently on accounting information when forming expansion plans.

In untabulated analyses, we split the informal sample by whether a family business tradition motivated entrepreneurship. In the intensive-margin analysis, digitized accounting practices are significantly associated with expansion plans only among entrepreneurs without a family tradition, consistent with accounting quality being more informative when managerial human capital is not inherited. In contrast, in the extensive-margin analysis, the marginal effects of digitized practices are similar across subsamples, suggesting that even entrepreneurs with family business backgrounds rely on accounting information when considering more substantial expansion decisions.

²⁸ After adjusting for sampling weights, *Digitized* is statistically significant at the 0.05 level for the intensive margin of business expansion plans (see column (3) of Table IA6 in the Internet Appendix).

²⁹ The reported 42 (30) percent effects are obtained by dividing the estimated coefficients in Table 10, column (4), for digitized records (0.135) and separate notebooks (0.096) by the sample mean of the indicator variable for informal firms with business expansion plans along the extensive margin (0.318).

Overall, while prior research has focused primarily on realized capital investment, we provide descriptive evidence on the role of accounting quality in the planning phase of capital budgeting, a necessary precondition for efficient investment. More broadly, our findings highlight the managerial role of accounting in informal microbusinesses, where formal reporting mandates are absent, but accounting is nevertheless associated with forward-looking decision-making (Datar et al., 2009).

5. Conclusion

This study provides new evidence on the voluntary adoption of accounting systems by microenterprises in the informal economy. In the absence of regulatory mandates, we find that informal firms adopt more structured recordkeeping practices when doing so aligns with their financial and operational objectives. These practices vary systematically with owner motivations, financing needs, and business strategies, consistent with rational economic behavior. Importantly, accounting quality is positively associated with access to credit, even among informal firms, suggesting that accounting practices may play a role in financial inclusion. Overall, our findings underscore the strategic role of accounting beyond compliance and highlight its potential to support microenterprise development. Future research could build on this work to better understand the evolution of accounting practices and credit relationships in informal settings.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Survey methodology

This section briefly summarizes the survey methodology based on ILO (2020). The survey was administered in mid-November 2019 to a sample of microbusinesses operating in Guatemala, Honduras, and El Salvador. The survey covered a wide range of topics, including general business and owner profiles as well as detailed information on operational and management practices such as strategic planning, human resource management, access to finance, recordkeeping, marketing strategies, and supply chain dynamics.

The survey targeted microbusinesses with fixed premises employing between two and ten permanent workers, including the owner (ILO, 2020). Data were collected through in-person interviews, mostly with the owners at their business premises (ILO, 2020).³⁰ Responses were recorded on electronic tablets using “SurveyGizmo” software, ensuring real-time capture and minimizing transcription errors. As part of quality control, the project coordinator accompanied the field team for at least 20 percent of the interviews to ensure that sampling filters were properly applied and that surveys were administered correctly. In addition, country-level research analysts downloaded and reviewed the collected data on a daily basis to check for consistency, completeness, and compliance with sampling quotas, using the SurveyGizmo platform to monitor incoming surveys in real time.

We do not observe the overall survey non-response rates, as this information is not included in the dataset. However, based on our review of the survey report prepared by the firm that administered the survey in 2019, overall response patterns do not appear to be a concern, as the report describes the dataset as “complete, well organized, and characterized by high response rates”.

To reduce response bias related to business practices and owner characteristics, questions on legal compliance were placed toward the end of the questionnaire. The survey includes questions that require answers on a 1–5 Likert scale, ranking-type questions (in which respondents selected their top three preferences from a given set of options), multiple-choice and yes or no questions, as well as open-ended numerical items (e.g., age). Data collection was administered by Research & Planning,³¹ a reputable market research firm with established operations across the three countries in our sample.

The survey was designed to ensure a representative cross-section of microenterprises and their characteristics, with observations evenly distributed across three domains. First, the geographic domain ensured balanced representation across the San Salvador Metropolitan area (El Salvador), the Tegucigalpa and Comayagua Metropolitan areas (Honduras), and Guatemala City (Guatemala). Second, the sectoral domain provided equal distribution among the retail sector (such as bars, restaurants, and hotels), the service sector (including professional, repair, beauty, health, education, and other personal services), and the micro-manufacturing sector (like tortilla shops, tailor shops, handicrafts, bakeries, and ice cream parlors). Finally, the sample is equally distributed across the three levels of microenterprise formalization: formal firms, sub-formal firms, and fully informal firms.

As discussed in the body of the paper, we follow the ILO (2020) classification and define formal microbusinesses as those registered with the national tax authority, indicating full compliance with business registration requirements and national laws and regulations. The rest are classified as informal microenterprises, consisting of sub-formal firms, i.e., those only locally registered in the public register of commerce (for Honduras and El Salvador) or the mayor’s office (for Guatemala) but lacking tax registration, and fully informal firms, which are not registered with any administrative authority. This parsimonious grouping is motivated by the lack of clear ex-ante predictions within the non-formal categories, allowing us to highlight key insights. Therefore, by design, informal firms represent two-thirds of the total sample, with formal firms comprising the remaining one-third.

³⁰ Only 6 percent of the survey was completed by managers with the owners (91 percent) and partners (3 percent) completing the rest.

³¹ This market research firm is also known as “Publicidad Comercial MullenLowe” (<https://www.pcomercial.com/>).

Because of the informal nature of many of the firms being studied, the sample was constructed without a sampling frame (a list of formal and informal businesses in the sectors of activity of interest within the needed geographic clusters). Therefore, a sample size calculation was used based on several assumptions. First, the required sample size is calculated under simple random sampling, and then the sample is evenly distributed among the previously described strata (three countries, three economic sectors, and three categories of formalization status). The formula used for the sample size is as follows:

$$n = \frac{z_{\alpha/2}^2 s^2}{e^2},$$

where n represents the total sample size, $z_{\alpha/2}$ is the two-tailed critical value of a standardized normal variable for a $1-\alpha$ confidence level, s^2 is the variance of the responses, and e is the maximum error allowed. Because the indicators of interest are ratios/proportions (e.g., the likelihood of adopting systematic recordkeeping or obtaining a bank loan), the survey design uses the maximum variance of a Bernoulli process (0.25) to calculate the sample size. A 95 percent confidence level and a maximum permissible error of 2.8 percent are used for the overall sample:

$$n = \frac{(1.96)^2(0.25)}{0.028^2} = 1225.$$

With the sample size rounded to the nearest integer that is divisible by 27, enabling equal distribution among the $3 \times 3 \times 3$ strata. This results in a total sample size of 1242 microenterprises with 46 firms per stratum to ensure the overall maximum allowed error does not deviate or exceed the target of 2.8 percent³²:

$$e = \frac{(1.96)(0.5)}{\sqrt{1,242}} = 0.02781$$

In other words, this maximum permissible error (e) refers to the largest acceptable difference between the sample estimate and the true population parameter. Therefore, for the full sample analysis (without dividing the sample by formality), our estimates should be within ± 2.8 percent of the true population values with 95 percent confidence. Following a similar calculation, when we analyze by formalization levels, the maximum permissible error is approximately 4.82 percent for the formal sector and 3.41 percent for the informal sector.

Actual data collection resulted in a slightly larger sample of 1426 firms than the required sample of 1242 firms. The designed sample allocation and the actual sample distribution across regions, levels of formalization, and sectors are provided below in Exhibit A1, except that we collapse the sub-formal and fully informal firms into one “Informal” category. Table A.1 presents a comparative descriptive analysis of formal and informal firms, while Table A.2 reports descriptive statistics for the main variables included in the accounting model. The observed patterns align with economic intuition and lend credibility to the reliability and representativeness of the survey data.

Exhibit A1.						
Sample distribution according to the survey design and effective sample						
Designed sample						
Sector	Guatemala		Honduras		El Salvador	
	Formal	Informal	Formal	Informal	Formal	Informal
Retail	46	92	46	92	46	92
Service	46	92	46	92	46	92
Micromanufacturing	46	92	46	92	46	92
Total	138	276	138	276	138	276
Effective sample						
Sector	Guatemala		Honduras		El Salvador	
	Formal	Informal	Formal	Informal	Formal	Informal
Retail	46	118	46	144	46	112
Service	46	91	46	100	46	98
Micromanufacturing	46	94	46	109	46	91
Total	138	303	138	353	138	301

³² ILO reported a denominator of 1224 with an e of 0.02801, while the denominator should have been 1242.

Table A.1
Means of determinants of accounting choice.

	Informal	Formal	Difference
Owner motivation			
<i>Business Orientation</i>	0.674	0.669	0.006
<i>Family Time</i>	0.667	0.656	0.011
<i>Barriers</i>	0.510	0.446	0.064***
Owner characteristics			
<i>Secondary School</i>	0.476	0.532	-0.056**
<i>College</i>	0.068	0.220	-0.152***
<i>Management Training</i>	0.521	0.722	-0.201***
<i>Female</i>	0.499	0.387	0.113***
<i>Age</i>	42.433	44.126	-1.693***
Financing			
<i>Saving for Investment</i>	0.460	0.493	-0.033***
<i>Own Capital</i>	0.528	0.513	0.015
<i>Informal Lenders</i>	0.464	0.254	0.210***
<i>Microfinance</i>	0.166	0.199	-0.033
<i>Bank</i>	0.269	0.524	-0.255***
Operations			
<i>Large Supplier</i>	0.211	0.406	-0.195***
<i>Customer Credit</i>	0.026	0.113	-0.087***
<i>Marketing Promotion</i>	0.583	0.929	-0.346***
<i>Business Complexity</i>	0.026	0.100	-0.074***
<i>Firm Age</i>	7.440	9.448	-2.008***
<i>Number of Family Workers</i>	0.829	0.859	-0.030
<i>Temporary Workers</i>	0.119	0.182	-0.063***
<i>Number of Workers</i>	2.955	3.938	-0.983***
Prefer Formalization	0.618	0.928	-0.309***

This table reports mean values of determinants of accounting choice by formalization. The final column reports differences in means and the results of two-sample mean t-tests, with standard errors adjusted for unequal variance between the two groups. Inferences are identical using a chi-squared test. *, **, and *** represent significance levels of 0.10, 0.05, and 0.01, two-tailed tests, respectively.

Table A.2
Descriptive statistics for the determinants of accounting choice by informal firms.

	N	Mean	SD	Median	Min	Max
<i>Business Orientation</i>	949	0.674	0.132	0.688	0.188	1.000
<i>Family Time</i>	950	0.667	0.191	0.750	0.000	1.000
<i>Barriers</i>	940	0.510	0.176	0.500	0.000	1.000
<i>Secondary School</i>	957	0.476	0.500	0.000	0.000	1.000
<i>College</i>	957	0.068	0.252	0.000	0.000	1.000
<i>Management Training</i>	957	0.521	0.500	1.000	0.000	1.000
<i>Female</i>	957	0.499	0.500	0.000	0.000	1.000
<i>Age</i>	957	42.433	10.913	42.000	18.000	81.000
<i>Saving for Investment</i>	949	0.460	0.218	0.500	0.000	1.000
<i>Own Capital</i>	957	0.528	0.499	1.000	0.000	1.000
<i>Informal Lenders</i>	957	0.464	0.499	0.000	0.000	1.000
<i>Microfinance</i>	957	0.166	0.372	0.000	0.000	1.000
<i>Bank</i>	957	0.269	0.443	0.000	0.000	1.000
<i>Large Supplier</i>	957	0.211	0.408	0.000	0.000	1.000
<i>Customer Credit</i>	957	0.026	0.160	0.000	0.000	1.000
<i>Marketing Promotion</i>	957	0.583	1.017	0.000	0.000	6.000
<i>Business Complexity</i>	957	0.026	0.160	0.000	0.000	1.000
<i>Firm Age</i>	957	7.440	7.645	5.000	0.000	60.000
<i>Number of Family Workers</i>	957	0.829	0.970	1.000	0.000	7.000
<i>Temporary Workers</i>	957	0.119	0.324	0.000	0.000	1.000
<i>Number of Workers</i>	957	2.955	1.284	3.000	2.000	9.000

Appendix B. Variable definitions

Unless otherwise indicated, all survey question numbers refer to the Central American survey.

Variable definitions for the accounting model

Variable name	Definition
Accounting variables	
<i>Systematic Recordkeeping</i>	Equals one when the business maintains formal accounting records and zero otherwise (Q24).
<i>Commingle NB</i>	Equals one when the business uses a notebook or notepad for keeping its formal accounting records (Q24 and Q25), and the entrepreneur does not separate business accounts from personal funds used to support their household (Q.26.1) and zero otherwise.
<i>Separate NB</i>	Equals one when the business uses a notebook or notepad for keeping its formal accounting records (Q24 and Q25), and the entrepreneur keeps separate business accounts from personal funds used to support their household (Q.26.1) and zero otherwise.
<i>Digitized</i>	Equals one when the business uses Excel spreadsheets, has an in-house accountant or bookkeeper, or outsources its accounting services for keeping its formal accounting records and zero otherwise (Q24 and Q25).
Owner motivation	
<i>Business Orientation</i>	The average of four measures (each on a 1–5 Likert scale) of possible motivations for starting the business, capturing the entrepreneur's business-oriented nature, standardized to a 0–1 scale. Specifically, it includes: (i) wanting to become independent and be one's own boss (Q10.3), (ii) continuing a family business tradition (Q10.4), (iii) having long aspired to own a business and seizing an opportunity when it arose (Q10.5), and (iv) viewing business ownership as the primary way to generate income (Q10.6).
<i>Family Time</i>	The average of two measures (each on a 1–5 Likert scale) of possible motivations for starting the business, capturing the entrepreneur's desire for flexibility (Q10.9) and balancing business/family (Q10.10), standardized to a 0–1 scale.
<i>Barriers</i>	The average of two measures (each on a 1–5 Likert scale) for starting the business due to a lack of employment opportunities (Q10.1) and labor market discrimination (Q10.11), standardized to a 0–1 scale.
Owner characteristics	
<i>Secondary School</i>	Denotes if the highest educational attainment of the entrepreneur is either middle school or high school completion (Q6.3).
<i>College</i>	Denotes if the highest educational attainment of the entrepreneur is completed college or higher technical education (Q6.4 and Q6.5).
<i>Management Training</i>	Equals one if the business owner has received training—through formal studies, training courses, or self-learning—in developing skills to manage staff and run the business; equals zero otherwise (Q16).
<i>Female</i>	This indicator variable identifies female business owners (Q4.2).
<i>Age</i>	Age of the entrepreneur in years (Q5).
Financing	
<i>Saving for Investment</i>	The entrepreneur's assessment of the likelihood of saving money for investment (measured on a 1–5 Likert scale, standardized to a 0–1 scale) (Q46.1).
<i>Own Capital</i>	Equals one when the entrepreneur views their own resources as a source they would consider for financing business expansion and zero otherwise (Q31.1).
<i>Informal Lenders</i>	Equals one when the entrepreneur views family and friends or individual lenders as a source they would consider for financing business expansion and zero otherwise (Q31.2 and Q31.3).
<i>Microfinance</i>	Equals one when the entrepreneur views microfinance companies as a source they would consider for financing business expansion and zero otherwise (Q31.4).
<i>Bank</i>	Equals one when the entrepreneur views private or public banks as a source they would consider for financing business expansion and zero otherwise (Q31.5 and Q31.6).
Operations	
<i>Large Supplier</i>	Equals one for entrepreneurs who have a large business as a supplier (Q38.8 and Q38.9).
<i>Customer Credit</i>	This indicator variable captures which businesses accept installment credit as a form of payment from their customers, whether from individuals or other businesses (Q45.2 and Q44.2).
<i>Marketing Promotion</i>	Sum of the indicator variables for six promotion strategies: newspaper advertisements, radio spots, mural advertisements, street marketing, flyer distribution, and social media engagement (Q37.2, Q37.3, Q37.4, Q37.5, Q37.6, Q37.10).

<i>Business Complexity</i>	Equals one when the business has additional establishments and zero otherwise (Q3.a).
<i>Firm Age</i>	The number of years the business has been operating (Q7).
<i>Number of Family Workers</i>	Number of family workers among the people working in the business (Q23).
<i>Temporary Workers</i>	Equals one when the business occasionally employs temporary workers (Q21).
<i>Number of Workers</i>	Number of permanent workers regularly employed in the business, including the owner (Q1).
Formalization variables	
<i>Stated Preference for Formalization</i>	Equals one for entrepreneurs who prefer operating with formal registration (Q49).
<i>Informal</i>	Equals one for businesses not registered with the tax authority and zero otherwise (Q54.4).
<i>Formal</i>	Equals one for businesses registered with the tax authority and zero otherwise (Q54.4).
Business cell phone use	Equals one for entrepreneurs who use a cell phone for business purposes, e.g., to receive orders or contact customers and suppliers (Q47.1.1, Q47.1.2, Q47.1.3, Q47.1.5).
Variable definitions for the loans model and credit status (not defined above)	
<i>Loan Requests</i>	Equals one if the business has ever requested financing from a bank and zero otherwise (Q33).
<i>Loan Approval</i>	Equals one (zero) if the business was (was not) granted a loan following a financing request to a bank (Q34).
<i>Credit Market Status</i>	Central American Sample: Equal to 3 when a loan was requested and obtained, 2 when no loan was requested and financing was not needed, 1 when a loan was requested but not obtained, and 0 when no loan was requested and financing was needed (Q32, Q33, Q34). Dominican Republic Sample: Equal to 3 when High Loan equals 1, 2 when High Loan equals 0, 1 when no loan was requested and financing was not needed, and 0 when no loan was requested and financing was needed. For the 10 observations where the loan request was rejected, we set the loan amount to 0 and High Loan to 0. (Q37, Q38, Q40).
<i>High Loan</i>	For the Dominican Republic sample, equals one when the loan amount obtained is greater than or equal to 100,000 Dominican pesos (Q38.30).
<i>Loan Amount</i>	For the Dominican Republic sample, refers to the amount of the loan received in Dominican pesos. It equals zero if the loan application was not approved (Q38.30).
<i>Owns Business Premise</i>	Equals one if the business premises are self-owned and zero otherwise (Q29).
<i>Lessee</i>	Equals one if the business is located in a shopping center, market, or building that is not the home of the owner, and zero otherwise (Q5e, Q29).
<i>Supplier Credit</i>	Equals one when the business uses term credit as the most common form of payment with its suppliers and zero otherwise (Q39).
<i>I Am Never Short of Money</i>	Equals one when the entrepreneur reports never being short of money for regular business operations and zero otherwise (Q32.1).
<i>Large Bank</i>	For the Dominican Republic sample, equals one if the last loan application was made to a large bank—Banreservas, Banco Popular, Banco BHD, Scotiabank, or Asociación Popular de Ahorros y Préstamos—and zero if the application was made to any other bank (Q39). The variable is coded only for businesses that reported having requested a loan.
Variable definitions for the business expansion model (not defined above)	
<i>Productivity</i>	Winsorized monthly sales (Q61) per permanent worker (Q1) (in U.S. dollars)
<i>Business Expansion</i>	Sum of five indicator variables corresponding to whether the owner reports plans to (i) increase sales, (ii) open a new location, (iii) hire additional workers, (iv) introduce new products, and (v) start a new business (Q11), with the indicators taking a value of one when the microenterprise is likely, very likely, or nearly certain to engage in the business expansion activity.
<i>Intensive Margin</i>	Sum of four indicators from above related to the existing enterprise: (i) increase sales, (ii) open a new location, (iii) hire additional workers, and (iv) introduce new products (Q11).
<i>Extensive Margin</i>	Equals one if the entrepreneur is likely, very likely, or nearly certain to start a new business distinct from the current one. (Q11.3).
<i>Log of Sales</i>	The natural logarithm of the estimated typical monthly sales, winsorized at the 1st and 99th percentiles prior to log transformation (Q61). Reported sales numbers in local currency are converted to U.S. dollars using the exchange rates at the time of the survey (mid-November 2019).

Appendix C. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jacceco.2026.101934>.

Data availability

The survey data used in this study were provided gratis by the International Labour Organization's Department of Employers' Activities (ACT/EMP), San José, Costa Rica.

References

- Aga, Gemechu Ayana, Reilly, Barry, 2011. Access to credit and informality among micro and small enterprises in Ethiopia. *Int. Rev. Appl. Econ.* 25 (3), 313–329.
- Ahmed, Anwer S., Duellman, Scott, 2013. Managerial overconfidence and accounting conservatism. *J. Account. Res.* 51 (1), 1–30.
- Allee, Kristian D., Yohn, Teri Lombardi, 2009. The demand for financial statements in an unregulated environment: An examination of the production and use of financial statements by privately held small businesses. *Account. Rev.* 84 (1), 1–25.
- Altonji, Joseph G., Elder, Todd E., Taber, Christopher R., 2005. Selection on observed and unobserved variables: Assessing the effectiveness of catholic schools. *J. Political Econ.* 113 (1), 151–184.
- Angelucci, Manuela, Karlan, Dean, Zinman, Jonathan, 2015. Microcredit impacts: Evidence from a randomized microcredit program placement experiment by compartamos Banco. *Am. Econ. J.: Appl. Econ.* 7 (1), 151–182.
- Aobdia, Daniel, 2019. Do practitioner assessments agree with academic proxies for audit quality? Evidence from PCAOB and internal inspections. *J. Account. Econ.* 67 (1), 144–174.
- Armstrong, Christopher S., Guay, Wayne R., Weber, Joseph P., 2010. The role of information and financial reporting in corporate governance and debt contracting. *J. Account. Econ.* 50 (2), 179–234.
- Augsburg, Britta, De Haas, Ralph, Harmgart, Heike, Meghir, Costas, 2015. The impacts of microcredit: Evidence from Bosnia and Herzegovina. *Am. Econ. J.: Appl. Econ.* 7 (1), 183–203.
- Ayyagari, Meghana, Demircuc-Kunt, Asli, Maksimovic, Vojislav, 2010. Firm innovation in emerging markets: The role of finance, governance, and competition. *J. Financ. Quant. Anal.* 46 (6), 1545–1580.
- Ball, Ray, 2024. By what criteria do we evaluate accounting? Some thoughts on economic welfare and the archival literature. *J. Account. Res.* 62 (1), 7–54.
- Ball, Ray, 2025a. Markets and the spontaneous emergence of double-entry accounting: A short essay. *Account. Horiz.* 39 (2), 1–6.
- Ball, Ray, 2025b. On The Economics of Accounting in Firms. The University of Chicago Booth School of Business.
- Ball, Ryan, Bushman, Robert M., Vasvari, Florin P., 2008. The debt-contracting value of accounting information and loan syndicate structure. *J. Account. Res.* 46 (2), 247–287.
- Banerjee, Abhijit V., Breza, Emily, Duflo, Esther, Kinnan, Cynthia, 2017. Do Credit Constraints Limit Entrepreneurship? Heterogeneity in the Returns to Microfinance. Working Paper 3126359, Working Paper, Rochester, NY.
- Banerjee, Abhijit V., Duflo, Esther, 2014. Do firms want to borrow more? Testing credit constraints using a directed lending program. *Rev. Econ. Stud.* 81 (2), 572–607.
- Banerjee, Abhijit, Duflo, Esther, Glennerster, Rachel, Kinnan, Cynthia, 2015. The miracle of microfinance? Evidence from a randomized evaluation. *Am. Econ. J.: Appl. Econ.* 7 (1), 22–53.
- Banerjee, Abhijit, Munshi, Kaivan, 2004. How efficiently is capital allocated? Evidence from the knitted garment industry in Tirupur. *Rev. Econ. Stud.* 71 (1), 19–42.
- Bari, Faisal, Malik, Kashif, Meki, Muhammad, Quinn, Simon, 2021. Asset-based microfinance for microenterprises: Evidence from Pakistan. *Am. Econ. Rev.*
- Basu, Sudipta, 2015. Is There Any Scientific Basis for Accounting? Implications for Practice, Research and Education. Working Paper 2649263, Working Paper, Rochester, NY.
- Basu, Sudipta, Kirk, Marcus, Waymire, Greg, 2009. Memory, transaction records, and the wealth of nations. *Account. Organ. Soc.* 34 (8), 895–917.
- Basu, Sudipta, Waymire, Gregory B., 2006. Recordkeeping and human evolution. *Account. Horiz.* 20 (3), 201–229.
- Beaman, Lori, Karlan, Dean, Thuysbaert, Bram, Udry, Christopher, 2023. Selection into credit markets: Evidence from agriculture in Mali. *Econometrica* 91 (5), 1595–1627.
- Beaver, Wh, 1968. Information content of annual earnings announcements. *J. Account. Res.* 6, 67–92.
- Beck, Thorsten, Demircuc-Kunt, Asli, Laeven, Luc, Levine, Ross, 2008. Finance, firm size, and growth. *J. Money Credit. Bank.* 40 (7), 1379–1405.
- Beck, Thorsten, Demircuc-Kunt, Asli, Martinez Peria, Maria Soledad, 2007. Reaching out: Access to and use of banking services across countries. *J. Financ. Econ.* 85 (1), 234–266.
- Beck, Thorsten, Levine, Ross, Loayza, Norman, 2000. Finance and the sources of growth. *J. Financ. Econ.* 58 (1), 261–300.
- Berger, Allen N., Miller, Nathan H., Petersen, Mitchell A., Rajan, Raghuram G., Stein, Jeremy C., 2005. Does function follow organizational form? Evidence from the lending practices of large and small banks. *J. Financ. Econ.* 76 (2), 237–269.
- Berger, Allen N., Udell, Gregory F., 1995. Relationship lending and lines of credit in small firm finance. *J. Bus.* 68 (3), 351–381.
- Bernard, Darren, Sutherland, Andrew, 2025. Private Firms and the Economic Role of Accounting: A Review of Empirical Research. Working Paper 5197461, Social Science Research Network, Rochester, NY.
- Besley, Timothy, Coate, Stephen, 1995. Group lending, repayment incentives and social collateral. *J. Dev. Econ.* 46 (1), 1–18.
- Beuselinck, Christof, Elfers, Ferdinand, Gassen, Joachim, Pierk, Jochen, 2023. Private firm accounting: the European reporting environment, data and research perspectives. *Account. Bus. Res.* 53 (1), 38–82.
- Biddle, Gary C., Hilary, Gilles, 2006. Accounting quality and firm-level capital investment. *Account. Rev.* 81 (5), 963–982.
- Biddle, Gary C., Hilary, Gilles, Verdi, Rodrigo S., 2009. How does financial reporting quality relate to investment efficiency? *J. Account. Econ.* 48 (2), 112–131.
- Bloom, Nicholas, Eifert, Benn, Mahajan, Aprajit, McKenzie, David, Roberts, John, 2013. Does management matter? Evidence from India. *Q. J. Econ.* 128 (1), 1–51.
- Bloom, Nicholas, Mahajan, Aprajit, McKenzie, David, Roberts, John, 2010. Why do firms in developing countries have low productivity? *Am. Econ. Rev.* 100 (2), 619–623.
- Bloom, Nicholas, Van Reenen, John, 2007. Measuring and explaining management practices across firms and countries*. *Q. J. Econ.* 122 (4), 1351–1408.
- Brickley, James A., Smith, Jr., Clifford W., Zimmerman, Jerold L., 2003. Corporate governance, ethics, and organizational architecture. *J. Appl. Corp. Financ.* 15 (3), 34–45.
- Brown, J. David, Earle, John S., 2017. Finance and growth at the firm level: Evidence from SBA loans. *J. Financ.* 72 (3), 1039–1080.

- Bruhn, Miriam, Hommes, M., Khanna, M., Singh, S., Sorokina, A., Wimpey, J.S., 2017. MSME finance gap : assessment of the shortfalls and opportunities in financing micro, small, and medium enterprises in emerging markets.
- Bruhn, Miriam, Karlan, Dean, Schoar, Antoinette, 2010. What capital is missing in developing countries? *Am. Econ. Rev.* 100 (2), 629–633.
- Buehn, Andreas, Schneider, Friedrich, 2012. Shadow economies around the world: novel insights, accepted knowledge, and new estimates. *Int. Tax Public Financ.* 19 (1), 139–171.
- Bushman, Robert M., Smith, Abbie J., 2001. Financial accounting information and corporate governance. *J. Account. Econ.* 32 (1), 237–333.
- Calderon, Gabriela, Iacovone, Leonardo, Juarez, Laura, 2017. Opportunity versus necessity: understanding the heterogeneity of female micro-entrepreneurs. *World Bank Econ. Rev.* 30 (Supplement 1), S86–S96.
- Capasso, Salvatore, Ohnsorge, Franziska, Yu, Shu, 2022. Informality and financial development: A literature review. *Manch. Sch.* 90 (5), 587–608.
- Carrizosa, Richard, Ryan, Stephen G., 2017. Borrower private information covenants and loan contract monitoring. *J. Account. Econ.* 64 (2), 313–339.
- Cassar, Gavin, 2009. Financial statement and projection preparation in start-up ventures. *Account. Rev.* 84 (1), 27–51.
- Cassar, Gavin, Ittner, Christopher D., Cavalluzzo, Ken S., 2015. Alternative information sources and information asymmetry reduction: Evidence from small business debt. *J. Account. Econ.* 59 (2), 242–263.
- Chein, Flávia, Silva, Uirá Mendes Vieira Da, 2014. Imperfeições no mercado de crédito e racionamento de crédito: uma análise para o setor informal no Brasil. *Nova Econ.* 24 (1), 103–122.
- Choudhary, Preeti, Merkley, Kenneth, Schipper, Katherine, 2022. The costs of waiving audit adjustments. *J. Account. Res.* 60 (5), 1813–1857.
- Christensen, Hans B., Nikolaev, Valeri V., 2012. Capital versus performance covenants in debt contracts. *J. Account. Res.* 50 (1), 75–116.
- Christensen, Hans B., Nikolaev, Valeri V., Wittenberg-Moerman, Regina, 2016. Accounting information in financial contracting: The incomplete contract theory perspective. *J. Account. Res.* 54 (2), 397–435.
- Costello, Anna M., Wittenberg-Moerman, Regina, 2011. The impact of financial reporting quality on debt contracting: Evidence from internal control weakness reports. *J. Account. Res.* 49 (1), 97–136.
- Cotler, Pablo, Woodruff, Christopher, 2008. The impact of short-term credit on microenterprises: Evidence from the *fincomun* - *bimbo* program in Mexico. *Econ. Dev. Cult. Chang.* 56 (4), 829–849.
- Crépon, Bruno, Devoto, Florencia, Dufló, Esther, Parienté, William, 2015. Estimating the impact of microcredit on those who take it up: Evidence from a randomized experiment in Morocco. *Am. Econ. J.: Appl. Econ.* 7 (1), 123–150.
- Dancer, Diane, Rammohan, Anu, Smith, Murray D., 2008. Infant mortality and child nutrition in Bangladesh. *Health Econ.* 17 (9), 1015–1035.
- Datar, Srikan, Epstein, Marc, Yuthas, Kristi, 2009. Management accounting and control: Lessons for and from the World's Tiniest Businesses. *Strateg. Financ.*
- Dechow, Patricia, Ge, Weili, Schrand, Catherine, 2010. Understanding earnings quality: A review of the proxies, their determinants and their consequences. *J. Account. Econ.* 50 (2), 344–401.
- Demirgüç-Kunt, Asli, Klapper, Leora F., Panos, Georgios A., 2011. Entrepreneurship in post-conflict transition. *Econ. Transit.* 19 (1), 27–78.
- Diamond, Douglas W., 1984. Financial intermediation and delegated monitoring. *Rev. Econ. Stud.* 51 (3), 393–414.
- Diamond, Douglas W., Verrecchia, Robert E., 1991. Disclosure, liquidity, and the cost of capital. *J. Financ.* 46 (4), 1325–1359.
- Dickhaut, John, Basu, Sudipta, McCabe, Kevin, Waymire, Greg, 2010. Neuroaccounting: Consilience between the biologically evolved brain and culturally evolved accounting principles. *Account. Horiz.* 24 (2), 221–255.
- Dou, Yiwei, 2020. The debt-contracting value of accounting numbers and financial covenant renegotiation. *Manag. Sci.* 66 (3), 1124–1148.
- Drexler, Alejandro, Fischer, Greg, Schoar, Antoinette, 2014. Keeping it simple: Financial literacy and rules of thumb. *Am. Econ. J. Appl. Econ.* 6 (2), 1–31.
- Dupas, Pascaline, Robinson, Jonathan, 2013. Savings constraints and microenterprise development: Evidence from a field experiment in Kenya. *Am. Econ. J.: Appl. Econ.* 5 (1), 163–192.
- Dyer, W Gibb, West, Barry, Peacock, Ian, Yamada, Spencer, Dyer, Jessie, 2016. Can the poor be trained to be entrepreneurs? the case of the academy for creating enterprise in Mexico. *J. Dev. Entrep.* 21 (02), 1650008.
- Dyreg, Scott D., Vashishtha, Rahul, Weber, Joseph, 2017. Direct evidence on the informational properties of earnings in loan contracts. *J. Account. Res.* 55 (2), 371–406.
- Estevão, João, Lopes, José Dias, Penela, Daniela, 2022. The importance of the business environment for the informal economy: Evidence from the doing business ranking. *Technol. Forecast. Soc. Change* 174, 121288.
- Farazi, Subika, 2014. Informal Firms and Financial Inclusion : Status and Determinants. Working Paper, World Bank, Washington, DC.
- Ferracuti, Elia, Stubben, Stephen R., 2019. The role of financial reporting in resolving uncertainty about corporate investment opportunities. *J. Account. Econ.* 68 (2), 101248.
- Gassen, Joachim, Muhn, Maximilian, 2024. Financial Transparency of Private Firms: Evidence from a Randomized Field Experiment. Working Paper 3290710, Working Paper, Rochester, NY.
- Guiso, Luigi, Sapienza, Paola, Zingales, Luigi, 2004. The role of social capital in financial development. *Am. Econ. Rev.* 94 (3), 526–556.
- Halvorson-Quevedo, Raundi, 1991. The growing potential of micro-enterprises. *OECD Obs.* 173, 7–11.
- Hasebe, Takuya, 2022. Endogenous models of binary choice outcomes: Copula-based maximum-likelihood estimation and treatment effects - Takuya Hasebe, 2022.
- Healy, Paul M., Palepu, Krishna G., 2001. Information asymmetry, corporate disclosure, and the capital markets: A review of the empirical disclosure literature. *J. Account. Econ.* 31 (1), 405–440.
- Hope, Ole-Kristian, Thomas, Wayne, Vyas, Dushyantkumar, 2009. Transparency, Ownership, and Financing Constraints in Private Firms. Working Paper.
- Hope, Ole-Kristian, Thomas, Wayne, Vyas, Dushyantkumar, 2011. Financial credibility, ownership, and financing constraints in private firms. *J. Int. Bus. Stud.* 42 (7), 935–957.
- Hope, Ole-Kristian, Vyas, Dushyantkumar, 2017. Private company finance and financial reporting. *Account. Bus. Res.* 47 (5), 506–537.
- IFC, International Finance, 2010. Scaling-Up SME Access to Financial Services in the Developing World. Working Paper, World Bank, Washington, DC.
- ILO, 2016. Role of Finance in Driving Formalization of Informal Enterprises. Working Paper.
- ILO, 2020. Encuesta de micronegocios en los países del norte del norte de centroamérica.
- Jayachandran, Seema, 2020. Microentrepreneurship in Developing Countries. Working Paper 3522310, Social Science Research Network, Rochester, NY.
- Kahsay, Girmachew, Zeleke, Girma, 2019. Factors affecting use of accounting records on small & micro enterprises (the case of Debre Birhan City, Ethiopia). *J. Invest. Manag.* 8 (1), 1.
- Karlan, Dean, Valdivia, Martin, 2011. Teaching entrepreneurship: Impact of business training on microfinance clients and institutions. *Rev. Econ. Stat.* 93 (2), 510–527.
- King, Robert G., Levine, Ross, 1993. Finance and growth: Schumpeter might be right*. *Q. J. Econ.* 108 (3), 717–737.
- Kota, Ina, 2007. Back to basics - microfinance: Banking for the poor - finance & development - june 2007.
- La Porta, Rafael, Shleifer, Andrei, 2014. Informality and development. *J. Econ. Perspect.* 28 (3), 109–126.
- Laux, Volker, Ray, Korok, 2020. Effects of accounting conservatism on investment efficiency and innovation. *J. Account. Econ.* 70 (1), 101319.
- Levenson, Alec R., Maloney, William F., 1998. The Informal Sector, Firm Dynamics and Institutional Participation. Working Paper 5180, Social Science Research Network, Rochester, NY.
- Lisowsky, Petro, Minnis, Michael, 2020. The silent majority: Private U.S. firms and financial reporting choices. *J. Account. Res.* 58 (3), 547–588.
- Luchman, Joseph N., 2021. Determining relative importance in stata using dominance analysis: domin and domme. *Stata J.* 21 (2), 510–538.

- McKenzie, David, 2010. Impact assessments in finance and private sector development: What have we learned and what should we learn? *World Bank Res. Obs.* 25 (2), 209–233.
- McKenzie, David, Woodruff, Christopher, 2014. What are we learning from business training and entrepreneurship evaluations around the developing world? *World Bank Res. Obs.* 29 (1), 48–82.
- McKenzie, David, Woodruff, Christopher, 2017. Business practices in small firms in developing countries. *Manag. Sci.* 63 (9), 2967–2981.
- Mel, Suresh, McKenzie, David, Woodruff, Christopher, 2010. Who are the microenterprise owners?: Evidence from Sri Lanka on tokman versus de soto. In: Lerner, Josh, Schoar, Antoinette (Eds.), *International Differences in Entrepreneurship*. University of Chicago Press, pp. 63–88.
- Mel, Suresh, McKenzie, David, Woodruff, Christopher, 2013. The demand for, and consequences of, formalization among informal firms in Sri Lanka. *Am. Econ. J.: Appl. Econ.* 5 (2), 122–150.
- Minnis, Michael, 2011. The value of financial statement verification in debt financing: Evidence from private U.S. firms. *J. Account. Res.* 49 (2), 457–506.
- Minnis, Michael, Sutherland, Andrew, 2017. Financial statements as monitoring mechanisms: Evidence from small commercial loans. *J. Account. Res.* 55 (1), 197–233.
- Morgan, Bruce, 2019. An Analysis of Marketing and Microenterprise: Does Marketing Have an Impact on the Sustainability of These Unique Organizations?. Working Paper, Working Paper.
- Myers, Stewart C., Majluf, Nicholas S., 1984. Corporate financing and investment decisions when firms have information that investors do not have. *J. Financ. Econ.* 13 (2), 187–221.
- Naidu, Dharmendra, Ranjeeni, Kumari, 2024. Is customers' financial reporting quality associated with suppliers' decision to contract? *Account. Rev.* 99 (6), 325–350.
- Najera Ruiz, Tonatiuh, Collazzo, Pablo, 2021. Determinants of the use of accounting systems in microenterprises: evidence from Chile. *J. Account. Emerg. Econ.* 11 (4), 632–650.
- Pearlman, Sarah, 2014. Can low returns to capital explain low formal credit use? Evidence from Ecuador. *J. Dev. Areas* 48 (1), 1–20.
- Petersen, Mitchell A., Rajan, Raghuram G., 1994. The benefits of lending relationships: Evidence from small business data. *J. Financ.* 49 (1), 3–37.
- Porto, Rafael Barreiros, Costa, Rafaela da Rocha, Watanabe, Eluiza Alberto de Moraes, 2017. The multilevel effect of marketing activities on sales, revenue and profitability in a micro-enterprise. *Rev. Bras. Gestão Neg.* 19, 432–452.
- Prijadi, Ruslan, Wulandari, Permata, Desiana, Putri Mega, Pinagara, Fajar Ayu, Novita, Maya, 2020. Financing needs of micro-enterprises along their evolution. *Int. J. Ethics Syst.* 36 (2), 263–284.
- Roychowdhury, Sugata, Shroff, Nemit, Verdi, Rodrigo S., 2019. The effects of financial reporting and disclosure on corporate investment: A review. *J. Account. Econ.* 68 (2), 101246.
- Schmandt-Besserat, Denise, 1992. *Before Writing, Vol. I: From Counting to Cuneiform*. University of Texas Press.
- Schrand, Catherine M., Zechman, Sarah L.C., 2012. Executive overconfidence and the slippery slope to financial misreporting. *J. Account. Econ.* 53 (1), 311–329.
- Shen, Yun, 2021. CEO characteristics: a review of influential publications and a research agenda. *Account. Financ.* 61 (1), 361–385.
- Shroff, Nemit, 2017. Corporate investment and changes in GAAP. *Rev. Account. Stud.* 22 (1), 1–63.
- Soto, Hernando De, 2000. *The Mystery of Capital: Why Capitalism Triumphs in the West and Fails Everywhere Else*. Basic Books.
- Sunder, Shyam, 1997. *Theory of Accounting and Control*. South-Western College.
- Tomy, Rimmy E., Wittenberg-Moerman, Regina, 2024. Community membership and reciprocity in lending: Evidence from informal markets. *J. Account. Econ.* 101697.
- Tomy, Rimmy E., Wittenberg-Moerman, Regina, 2025. Information preference and credit allocation in a bazaar economy. *Manag. Sci.* 71 (12), 10642–10670.
- Ulysea, Gabriel, 2020. Informality: Causes and consequences for development. *Annu. Rev. Econ.* 12 (1), 525–546.
- Watts, Ross L., Zimmerman, Jerold, 1986. *Positive Accounting Theory*. Prentice Hall International.
- Waymire, Gregory B., Basu, Sudipta, 2008. Accounting is an Evolved Economic Institution. Working Paper 1155420, Working Paper, Rochester, NY.
- Weber, Michaela, McKenzie, David, 2009. The results of a pilot financial literacy and business planning training program for women in Uganda. *Financ. PSD Impact*.
- Wellalage, Nirosha Hewa, Locke, Stuart, 2016. Informality and credit constraints: evidence from Sub-Saharan African MSEs. *Appl. Econ.* 48 (29), 2756–2770.
- Williams, Colin C., Shahid, Muhammad S., Martínez, Alvaro, 2016. Determinants of the level of informality of informal micro-enterprises: Some evidence from the City of Lahore, Pakistan. *World Dev.* 84, 312–325.